The Rise of Red Equity

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Abstract

Empirical research routinely excludes firms with negative book equity, which we call red equity firms. Their share of U.S. listed firms rose from zero in the early 1960s to more than 10% in 2025, when their combined market value reached $2.6 trillion. The incidence of negative equity depends heavily on accounting treatment: consistent capitalization or expensing of intangibles would leave 2.5% or 31% of listed firms with negative equity, respectively. Classifying these firms by the source of their deficit reveals two distinct groups. (i) Payout-driven firms are mature and profitable; since 2010, they account for 87% of red equity firms' market value. Conditional on standard determinants, their leverage is positively associated with profitability, reversing the relation among positive-equity firms. (ii) Operating-loss-driven firms are younger, R&D-intensive, and mostly distressed. Including red equity firms raises the CAPM alpha of the intangible-adjusted value factor by more than one percentage point annually since 2009. Negative equity is similarly prevalent in the United Kingdom, France, and Australia. Researchers should therefore reconsider exclusions based on reported book equity and assess whether their conclusions persist when intangible capital is recognized.

Keywords: Negative equity, intangible assets, share repurchases, leverage, value relevance

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1 Introduction

Book equity is a central measure in accounting and finance. It measures shareholders' interest in a firm's recognized net assets, enters valuation and financial ratios such as the book-to-market ratio and the return on equity, and is often used as a proxy for the value recoverable in liquidation (Collins et al., 1999a). Its role extends beyond measurement: private loan agreements have commonly included covenants that require the borrower to maintain a minimum net worth, defined as assets minus liabilities (Demerjian, 2011), and both the Nasdaq and the New York Stock Exchange use stockholders' equity, together with market value, to decide whether a firm remains listed. Negative book equity is commonly read as a sign of financial distress (Fama and French, 1992; Griffin and Lemmon, 2002; Vassalou and Xing, 2004); the bankruptcy model of Ohlson (1980), for example, includes an indicator for total liabilities exceeding total assets. It also complicates the interpretation of standard ratios: return on equity, for example, becomes positive when both earnings and equity are negative.

Firms with negative book equity—hereafter red equity (RE) firms—are therefore routinely excluded from finance and accounting research (e.g., Fama and French, 1993, 2002; Strebulaev and Yang, 2013; Hou et al., 2020). Fama and French (1993) exclude them, noting that they are "rare before 1980" (p. 8), and the practice has since become standard: Hou et al. (2020), for example, exclude RE firms from the entire sample of anomalies they replicate. Other studies remove RE firms implicitly, through restrictions on related variables: Baker and Wurgler (2002) drop firm-years with book leverage above one, which, under their definition of book debt, excludes every firm-year with negative book equity, and Nissim and Penman (2001) exclude ratios with negative denominators, such as the return on common equity of RE firms. Even when RE firms are retained, their book-to-market ratios and returns on equity lose their usual interpretation, so that how they are treated can change the results.

We document that RE firms are no longer rare. Over the past decades, an increasing

number of well-known and profitable companies have negative book equity. Prominent examples in 2025 (with the number of years in which their common equity is negative in our sample in parentheses) include AbbVie (3), Philip Morris International (14), McDonald's (10), Booking Holdings (4), Lowe's (5), HCA Healthcare (14), McKesson (6), Altria (5), Starbucks (7), Marriott (8), TransDigm (13), AutoZone (17), Hilton (7), Yum! Brands (15), Dell Technologies (7), HP (10), and Domino's Pizza (22). All of them reported positive net income in 2025. This paper documents how common RE firms have become, explains why, and examines what their rise implies for research that relies on book equity.

Our central observation is that the number of RE firms is to a large extent an accounting outcome. U.S. GAAP capitalizes intangibles that a firm acquires, as identifiable intangible assets or goodwill, but expenses the same intangibles when the firm develops them internally. Applying either rule consistently changes the picture substantially (Figure 2). If acquired intangibles were expensed like internally generated ones, 31% of listed firms would have had negative book equity in 2025, three times the actual share, and these firms would have represented 27% of the market value of all listed firms (43% at its peak in 2019). If internally generated intangibles were instead capitalized like acquired ones, only 2.5% of firms would have had negative equity. We note that the first counterfactual uses only reported balance sheets and requires no estimate of intangible capital. Therefore, the sign of book equity, which researchers routinely use to screen firms, depends largely on whether a firm's intangibles were bought or built.

The rise of RE firms is substantial by any measure. The share of RE firms among U.S. listed firms, excluding utilities, banks, and insurers, grew from zero before 1962 to 4.7% in 2010 and to 11.5% in 2024 (10.5% in 2025). Of the 19,264 firms in our sample, 4,125, or 21%, have negative book equity in at least one year. The combined market value of RE firms reached a record $2.6 trillion in 2025, as much as that of the bottom 78% of listed firms ranked by market value, and the deficits have deepened: the median equity-to-assets ratio of RE firms fell from -0.11 in 1975 to -0.32 in 2024 (-0.27 in 2025).

To explain the rise, we start from the accounting identity. Book equity equals contributed capital plus cumulative earnings minus cumulative distributions (plus accumulated other comprehensive income), so it turns negative mainly through losses or through payouts to shareholders.[1] Losses, in turn, arise from three sources: operating losses in the economic sense, when costs exceed revenues; write-downs, which recognize at once a past loss in the value of acquired assets; and, under U.S. GAAP, the expensing of internally generated intangible investment, which consists of R&D and the investment part of selling, general, and administrative expenses and reduces reported earnings even when it creates value. We measure the importance of each channel by reconstructing book equity with each one added back. Since 1995, adding back cumulative write-downs would remove 20% of RE firm-years, adding back cumulative payouts 31%, and capitalizing intangible investment 71%; together, the adjustments remove 85%. The expensing of intangible investment is thus the most important single channel.

These counterfactuals do not tell us what kind of firm an RE firm is, because capitalizing intangibles turns most RE firm-years positive regardless of where the deficit comes from. We therefore classify RE firms into two groups, payout-driven and loss-driven, by the source of their deficit, using lifetime earnings, defined as retained earnings in the first year in which the firm reports them, plus the dividends paid up to that year and all net income since. Payout-driven RE firms have non-negative lifetime earnings: over their recorded history they have earned at least as much as they have lost, so their deficit reflects distributions to shareholders. Loss-driven RE firms have negative lifetime earnings, from operating losses or from write-downs. The two groups are very different. Since 2010, payout-driven firms account for 24% of RE firm-years but 87% of the market value of RE firms. They are large, mature (listed for a median of 22 years), and profitable: 78% report positive net income in a given year, against 57% of other firms. Loss-driven firms are the mirror image. They

[1] Accumulated other comprehensive income can also contribute: among RE firm-years since 1998 with accumulated other comprehensive income reported, 7% would have non-negative equity without it, and 20% of payout-driven ones.

account for 76% of RE firm-years but only 13% of RE market value, and only 20% of them are profitable. Most of them owe their deficit to operating losses rather than write-downs (71% of RE firm-years).[2]

The distinction matters for how negative equity should be interpreted. As a group, RE firms are often distressed: over 1975–2025, 69% of RE firm-years are classified as distressed by standard measures, against 15% of other firm-years, and 23% are followed by a performance-related delisting within two years, against 4%. This average, however, mixes two groups. Since 2010, 79% of the firm-years of firms whose deficit comes from operating losses are distressed, and 18% are followed by a delisting within two years. Payout-driven RE firms, by contrast, are not more distressed than the average listed firm: since 2010, 16% of their firm-years are classified as distressed, against 19%. Their negative equity reflects distributions financed largely with debt rather than accumulated losses. Leverage regressions are consistent with this interpretation. Among firms with positive equity, more profitable firms have lower book leverage, as predicted by the pecking-order theory; among payout-driven RE firms, more profitable firms have higher book leverage, consistent with the trade-off theory and with these firms borrowing against their earnings to fund distributions.

Negative equity also changes the meaning of book value for valuation. In regressions of the share price on book value and earnings per share, book value is positively priced for firms with positive equity (1.14) but negatively for RE firms, most strongly for payout-driven firms (-3.59, against -1.83 for loss-driven firms), whose market value rests on their earnings. Once intangible capital is added to book equity, book value is positively priced for RE firms of both types, and for payout-driven firms at a rate statistically indistinguishable from that of firms with positive equity. The treatment of RE firms matters for asset pricing as well. Including RE firms in an intangible-adjusted value factor raises its average return by 0.47 percentage points per year and its CAPM alpha by 0.57 percentage points over 1970–2026. Since 2009, the increases are almost twice as large, at 0.82 and 1.18 percentage points, and

[2]Their losses partly reflect expensed R&D, but for most of them cumulative losses exceed all the intangible investment they have ever expensed.

they rise steadily the later the sample starts. Because RE firms have become common only recently, and the payouts and intangible investment that create them continue to grow, the consequences of excluding them are likely to increase.

Negative book equity is not only a U.S. phenomenon. We find that RE firms are almost as common in the United Kingdom (9.4% of listed firms in 2025), France (10.6%), and Australia (9.1%) as in the United States, but almost absent in Japan, Korea, and China. Outside the United States, negative equity results less often from payouts. Pooling all listed firms incorporated in the 27 EU member states, the United Kingdom, Switzerland, Norway, and Iceland, which we refer to as Europe, 11% of RE firm-years over 2015–2025 are payout-driven, against 24% in the United States. The accounting result also holds abroad: in this pooled European sample, expensing all intangible assets on the balance sheet, including goodwill, would raise the share of listed firms with negative equity from 6% to 24% in 2025.[3]

Our results have direct implications for empirical research and for standard setting. Excluding RE firms, or treating negative equity as a sign of distress, mixes a financing choice with accumulated losses and removes some of the largest firms in the economy from the sample. Classifying RE firms by the source of the deficit, which requires only Compustat data, separates healthy from fragile firms. For standard setters, our results show that the asymmetric treatment of acquired and internally generated intangibles largely determines how many firms report negative equity, a point that bears directly on the current debate on accounting for intangibles (Lev and Gu, 2016; Lev, 2019; Penman, 2024; Financial Accounting Standards Board, 2024). A consistent treatment in either direction would change significantly the number of firms with negative equity, but neither would make its sign a reliable indicator of a firm's condition. Expensing acquired intangibles would add mostly large, profitable acquirers. Capitalizing internally generated intangibles would shrink the group to less than a third of its current size, but would leave the same two kinds of firms in

[3]India provides a further illustration of the role of reporting rules: when a 2011 reform required accumulated losses previously shown as assets to be deducted from shareholders' funds, the share of Indian firms with negative equity rose from 0.6% to 7.1% in a single year.

it: loss-driven firms, which would remain the majority, and payout-driven firms, which would still account for most of its market value. In practice, intangible-adjusted book equity offers a simple way to keep RE firms in empirical work. These firms are excluded not because they are distressed but because negative book equity makes ratios such as the book-to-market ratio or the return on equity meaningless. Adding intangible capital restores positive equity, and with it meaningful ratios, for about 70% of RE firm-years, and the resulting book value is positively priced for RE firms of both types.

To the best of our knowledge, our study provides the first comprehensive analysis of RE firms, documenting their sharp increase over the past two decades, examining the drivers of this trend, and assessing some of its implications for the empirical literature. Searching the 1,822 business-related journals included in the 2024 edition of the Chartered Association of Business Schools' Academic Journal Guide (AJG)[4], we identified only two papers that focus on RE firms, both using data ending in the mid-2000s. Specifically, using data up to 2004, Brown et al. (2008) classify RE firms as either value or growth stocks—based on past return characteristics—and find that most are categorized as value firms. They attribute this to the financial distress commonly associated with RE firms. Similarly, Jan and Ou (2012), using data up to 2005, find that RE firms tend to have higher market valuations per dollar of assets than firms with positive book equity. Our results also relate to the literature on the value relevance of book value and earnings (Collins et al., 1999b; Barth et al., 2023) and on losses driven by the expensing of intangibles (Gu et al., 2023).

A growing literature constructs intangible-adjusted measures and factors, using a variety of methodologies and parameter choices, and shows that incorporating intangibles improves the informational content of standard asset-pricing factors (Eisfeldt and Papanikolaou, 2013; Eisfeldt et al., 2022; Park, 2022; Jagannathan et al., 2023; Gulen et al., 2025; Iqbal et al., 2025; Ewens et al., 2025). We show that the set of RE firms is much smaller when book equity is adjusted for intangibles, and that how RE firms are treated has a significant and

[4]The journal list spans a broad range of disciplines, encompassing more than 115 journals in finance alone.

growing effect on the intangible-adjusted value factor.

Our paper is also related to the literature on missing financial data and the resulting exclusion of observations in empirical studies (Koh and Reeb, 2015; Blanchet et al., 2022; Beckmeyer and Wiedemann, 2022; Freyberger et al., 2024; Chen and McCoy, 2024; Bryzgalova et al., 2024). While this literature typically examines a wide range of variables and proposes methods to impute missing values, we focus on a single but critical variable, book equity, and on the common practice of excluding firms for which it is negative.

The remainder of the paper is organized as follows. Section 2 describes the data and the main measures. Section 3 documents the rise of RE firms and Section 4 its drivers. Section 5 examines distress, Section 6 leverage, and Section 7 the implications for accounting, valuation, and asset pricing. Section 8 presents international evidence, and Section 9 concludes.

2 Data and measurement

Accounting data come from the CRSP/Compustat Merged Database for fiscal years 1950–2025 and market data from CRSP. We keep U.S.-incorporated firms with ordinary common shares listed on the NYSE, Amex, or Nasdaq and exclude utilities, banks, and insurance companies (Fama–French 49 industries 31, 45, and 46).

Book equity attributable to common shareholders is stockholders' equity (SEQ) minus preferred stock (redemption, liquidating, or par value, in that order). When SEQ is missing, we use total assets minus total liabilities and, in that case, also deduct minority interest, which would otherwise be counted as common equity; when both are missing, we use common equity (CEQ). *RedEquity* equals one if book equity is negative. Continuous variables are winsorized at 1% in each tail. The final sample has 223,283 firm-years.

To reconstruct book equity under the capitalization of internally generated intangibles, we add off-balance-sheet intangible capital to common equity. Measuring internally gen-

erated intangible capital remains an active area of research, with a variety of empirical approaches (e.g., Eisfeldt and Papanikolaou, 2013; Peters and Taylor, 2017; Eisfeldt et al., 2022; Boustanifar and Kang, 2022; Belo et al., 2022; Falato et al., 2022; Jagannathan et al., 2023; Boustanifar and Kang, 2024; Iqbal et al., 2025; Ewens et al., 2025). In this paper, as our main measure, we follow Peters and Taylor (2017) and proxy intangible investment as the sum of R&D expenditures and 30% of SG&A, capturing investments in innovation, brand-building, human capital, and organizational capital. Because the WRDS series ends in 2023, we extend it to 2025 by rolling each firm's WRDS stocks forward with the same accumulation rules, and we use our own replication of their method only for firms that are not in the WRDS series (Appendix A). Rolling the 2022 stocks forward in the same way reproduces the WRDS 2023 stocks within 1% for 98% of firms, and our replication reproduces the WRDS stocks within 5% for 91% of firm-years over 1950–2023. We show that our main results are similar with alternative measures of capitalized intangible investments of Ewens et al. (2025), Eisfeldt and Papanikolaou (2013), and Iqbal et al. (2025).

To identify the source of a firm's deficit, we use the fact that book equity equals contributed capital plus lifetime earnings minus lifetime distributions (dividends and share repurchases), plus other comprehensive income. We measure a firm's lifetime earnings as its retained earnings in the first year in which it reports them, plus the dividends paid in any earlier years, plus the sum of its net income since, using Compustat Fundamentals Annual so that the history starts with the firm's first record.[5] We call an RE firm-year *payout-driven* if measured lifetime earnings are non-negative: over its recorded history the firm has earned at least as much as it has lost, so its deficit reflects distributions to shareholders. We call it *write-down-driven* if lifetime earnings are negative but become non-negative once cumulative

[5]About a quarter of RE firm-years belong to firms whose first Compustat record does not report retained earnings. Starting from the first reported value, rather than from zero, ensures that earnings and losses before that record are counted; the dividends paid in the starting year and in the earlier records are added back because they reduced its retained earnings. Distributions and other equity adjustments before the first Compustat record cannot be observed, so measured lifetime earnings are a reconstruction from the available history.

write-downs (goodwill impairments and other pre-tax write-downs, available from 1995) are added back, and *operating-loss-driven* otherwise.[6] Because the write-down-driven group is small (100 firms), the regression analyses combine it with the operating-loss-driven group into *loss-driven* RE firms.[7]

We measure distress with three indicators and their combination. The first is an Altman Z-score below 1.81 (Altman, 1968). The second is a naive default probability above 10% (Bharath and Shumway, 2008): a simplified version of the Merton model in which the firm defaults if, within one year, the value of its assets falls below a default point equal to short-term debt plus half of long-term debt; asset value is the market value of equity plus this debt, and its volatility is approximated from the stock's return volatility. The third is EBITDA below interest expense. A firm is *distressed* if at least two of the three hold. The Z-score and the interest-coverage indicator are (largely) accounting-based: they penalize firms that invest in intangibles, because EBIT and EBITDA are reported net of expensed intangible investment and retained earnings are correspondingly low. The default probability is market-based and does not depend on how intangibles are accounted for. As a realized outcome distinct from the accounting-based measures, we use performance-related CRSP delistings within 24 months of the fiscal year end (Shumway, 1997).[8]

We measure firm age, *Age*, as the number of years since the firm's initial public offering: the fiscal year minus the IPO year, set to zero for fiscal years before the IPO. The IPO year is the year of the firm's first public offering in Jay Ritter's IPO database, matched through

[6] Write-downs are added back at their pre-tax amounts, here and in the counterfactuals of Section 4.2. Because part of a write-down is offset by its tax effect, this overstates its effect on book equity: for RE firms, cumulative after-tax write-downs in Compustat are 72% of the pre-tax amounts, and with after-tax amounts the write-down-driven group is about two fifths smaller. The regressions, which pool write-down- and operating-loss-driven firms, are not affected.

[7] We note that a simpler classification based on the sign of retained earnings would not work: retained earnings are reduced by dividends and, for firms that retire repurchased shares, by repurchases, so that 73% of payout-driven firm-years have negative retained earnings.

[8] These include delistings for failing exchange equity requirements (CRSP reason EQRQ), which can depend on reported equity. They account for 2.5% of performance delistings, and excluding them leaves the delisting rates and the comparisons in Section 5 essentially unchanged.

the CRSP PERMNO. For firms that the database does not cover, including all firms that went public before 1975, we use the IPO date reported in Compustat and, if that is also missing, the first fiscal year in which the firm appears in Compustat; for firms listed before Compustat coverage begins, this last fallback understates age. *Age* enters the regressions of RE status in years and the leverage regressions as $\log(1 + Age)$.

Table 1 reports descriptive statistics, and Appendix Table A1 defines all variables.

3 The rise of red equity

We begin by documenting how common RE firms have become and how much their number depends on the accounting treatment of intangibles. If RE firms were still rare, as Fama and French (1993) observed, excluding them would be innocuous; the evidence in this section shows that this premise no longer holds.

Figure 1 plots the share of RE firms among listed firms from 1950 to 2025, and Table 2 reports their number, share, and market value by year. There were no RE firms before 1962. Their share then rose gradually to about 5% by 2010 and accelerated over the past decade: in 2025, more than one in ten listed firms had negative book equity (10.5%, after a peak of 11.5% in 2024). Their combined market value reached $2.6 trillion in 2025, as much as that of the bottom 78% of listed firms ranked by market value, and they accounted for 8.6% of the employment of all listed firms.

How many RE firms there are depends heavily on how intangibles are accounted for. Current GAAP is a hybrid: intangibles are capitalized when acquired and expensed when developed internally. Two consistent treatments bound it. If internally generated intangibles were also capitalized, book equity would be common equity plus intangible capital. If acquired intangibles were also expensed, book equity would be common equity minus the intangible assets on the balance sheet (including goodwill), that is, tangible equity.[9] Figure 2

[9]Subtracting the carrying value of acquired intangibles is equivalent to expensing them at acquisition, because their past amortization and impairments have already reduced common equity.

plots the share of RE firms under the three treatments, for the same firm-years. Under consistent expensing, the share of RE firms rises from 2.3% in 1975 to 13.9% in 2000, 17.2% in 2010, and 30.6% in 2025, against 1.2%, 5.4%, 4.7%, and 10.5% under current GAAP; under consistent capitalization, it never exceeds 3%.[10] Measured by market value, the contrast is even sharper: firms with negative tangible equity represent 27.2% of the market in 2025 and 43.5% at the 2019 peak, against 4.4% and 6.2% for RE firms under current GAAP (Figure 2, panel b). Unlike the capitalization counterfactual, the expensing counterfactual requires no estimate of intangible capital, only reported balance-sheet data.

The gap between current GAAP and consistent expensing widened from 8.5 percentage points in 2000 to 12.5 in 2010 and 20 in 2025. Two changes in the accounting for business combinations contributed to this widening. SFAS 141 (2001) abolished the pooling-of-interests method, under which acquisitions created no goodwill, and SFAS 142 (effective 2002) replaced the amortization of goodwill with impairment testing, so that goodwill now remains on the balance sheet until it is impaired. Both increase the acquired intangibles that GAAP recognizes, and therefore the number of firms whose book equity is positive only because their intangibles were bought.

We document four further facts about RE firms. First, their deficits have deepened: the median equity-to-assets ratio of RE firms fell from -0.11 in 1975 to -0.32 in 2024 and -0.27 in 2025 (Figure 3). Second, a substantial minority of RE firms are large: between 26% and 42% of RE firms were larger than the median listed firm in each year of 2020–2025 (Figure 4). Third, they include many mature firms: over the full sample, about 40% of RE firm-years belong to firms listed for ten years or more (Figure 5). Fourth, RE firms are present in all major industries, with the highest shares in healthcare and high technology (Figure 6).

[10] Compustat reports goodwill separately from other intangible assets only from fiscal year 1988; before 1988, goodwill is included in total intangible assets and cannot be distinguished from them. Goodwill accounted for about half of the intangible assets on the balance sheets of our sample firms in the 1990s and for between 60% and 69% in every year since 2002. Appendix Figure A3 shows that deducting only goodwill from common equity, or only the other intangible assets, produces the same upward trend in the share of firms with negative equity as deducting all intangible assets, at lower levels.

In sum, RE firms are no longer rare: more than one in ten listed firms has negative book equity, these firms are worth trillions of dollars, and they are not confined to small or young firms. At the same time, how many firms have negative equity depends to a large extent on how intangibles are accounted for, a point we return to throughout the paper.

4 What drives red equity?

Section 3 showed that RE firms have become common and that their number depends heavily on the accounting treatment of intangibles. This section asks what drives negative equity under current rules. We first use the accounting identity to define the channels through which book equity can turn negative (Section 4.1), then measure how much each channel contributes, both mechanically and in regressions (Section 4.2), ask whether the rise reflects changing firm characteristics (Section 4.3), and finally classify RE firms by the source of their deficit (Section 4.4).

4.1 Four accounting channels

Book equity changes through earnings and through transactions with shareholders. It therefore turns negative mainly through losses or through distributions in excess of earnings and contributed capital. Losses, in turn, arise from operating losses, from write-downs, and, because GAAP expenses internally generated intangible investment, from the expensing of investment that may create value. We treat these as four channels.

The first channel is losses. The share of firms reporting negative net income rose from 16% in 1970 to 52% in 2025 (Figure 7). A large part of these losses reflects the expensing of intangible investment: after adding back after-tax intangible investment, between 36% and 55% of loss-making firms would report a profit, depending on the year (Figure 8).

The second channel is payouts, which reduce book equity at their market value. Following the adoption of SEC Rule 10b-18 in 1982, which reduced the legal and operational frictions

associated with open-market repurchases, buybacks became an increasingly prominent payout channel: the share of repurchasing firms rose from 23% in 1971 to 64% in 2025, while the share of dividend payers fell from 60% to 29% (Figure 9, panel a). Aggregate payouts relative to assets doubled, from 2.5% to 4.9%, entirely because of repurchases, which rose from 0.2% to 3.1% of assets, while dividends fell slightly, from 2.3% to 1.8% (panel b). For a firm whose market value far exceeds its book value, a buyback equal to a modest fraction of its market value can exhaust its book equity.

The third channel is write-downs. Impairments reduce assets and equity immediately, while later recoveries are not recognized. Write-downs are large and lumpy and concentrate in downturns, peaking after the dot-com bust in 2001–2002 and in the 2008 financial crisis (Figure 10).[11] Their importance has also grown with the accounting for goodwill: since SFAS 142, effective for fiscal years beginning after 15 December 2001, goodwill is no longer amortized and leaves the balance sheet only through impairments. Most write-downs are goodwill impairments (73% of the dollar amount since 1995), so this channel largely reflects the later recognition of value losses on acquired intangibles. Unlike the expensing of internal investment, which reduces equity whether or not the investment creates value, an impairment is recorded only when the acquired business is found to be worth less than its book value. Individual impairments can be very large: General Electric recorded a $22 billion goodwill impairment in 2018 related to its Power division, AOL Time Warner a $45.5 billion impairment charge in 2002 following its 2000 merger, Hewlett-Packard an $8.8 billion impairment in 2012, and Kraft Heinz a $15.4 billion impairment announced in 2019. Such charges can erode book equity quickly, especially among firms that have grown through acquisitions.

The fourth channel is the expensing of internally generated intangible investment (R&D, product design, marketing, organizational development, and human and brand capital),

[11] Compustat reports write-downs (items WDP and GDWLIP) from 1995, but before 2000 only a few firms report them: between 37 and 64 firms per year in our sample over 1996–1999, compared with about 1,200 per year in 2001 and 2002, when coverage becomes systematic. The low values before 2001 in Figure 10, and the small write-downs added back for those years in the classification of red equity firms and in the counterfactuals, therefore reflect limited reporting rather than an absence of write-downs.

which has grown much faster than tangible investment. Aggregate intangible investment relative to capital expenditure rose from about 0.5 in the 1960s to between 1.3 and 1.6 since 2019, and the average ratio of intangible investment to tangible assets rose from below 0.1 to about 0.3 (Figure 11). The same intangibles are treated differently when they are acquired: an acquired intangible asset, or the goodwill of an acquired business, is recognized on the balance sheet and leaves book equity unchanged. Whether a firm's intangible investment reduces its book equity therefore depends on whether the firm builds or buys its intangibles. All four channels have grown substantially over the past decades: more firms report losses, more firms repurchase shares and in larger amounts, write-downs have become larger, and intangible investment has outgrown tangible investment. The next subsection measures how much each of them contributes to negative equity.

4.2 How much does each channel matter?

We measure the contribution of each channel in two ways. The first is mechanical: we reconstruct each firm's book equity with one channel added back and count how many RE firm-years would turn positive. Figure 12 plots the share of firms that would remain RE firms after each adjustment, in the same units as Figure 1. Since 1995, when write-down data become available, adding back cumulative write-downs removes 20% of RE firm-years, adding back cumulative payouts 31% (repurchases alone 25%), and capitalizing intangible investment 71%. With all adjustments together, 85% of RE firm-years would have positive equity; the remaining 15% have negative equity that none of these channels explains, so that their deficit reflects losses beyond the expensing of intangible investment. The ranking of the channels is the same with the other intangible capital measures, whether we use only the firm-years each measure covers or all firm-years (Appendix Figure A1 and Figure A2).

The counterfactuals are informative about magnitudes but they adjust one channel at a time and ignore the other characteristics of RE firms. Our second approach is therefore a regression of RE status on the variables that correspond to the four channels, together with

firm characteristics. We estimate linear probability models

$$RedEquity_{i,t} \times 100 = \alpha_t + \alpha_j + \beta' C_{i,t} + \theta' Z_{i,t} + \varepsilon_{i,t},$$

where $RedEquity_{i,t}$ equals one if firm i has negative book equity in year t, and $C_{i,t}$ are the channel variables: an indicator for a net loss ($LossMaking$), profitability ($Profitability$, operating income before depreciation over total assets), intangible investment over total assets ($Intan_over_Assets$), payouts, dividends plus repurchases, over total assets ($Payout_over_Assets$), and write-downs over total assets ($Writedown_over_Assets$). The controls $Z_{i,t}$ are size, leverage, age, and Tobin's Q, and α_t and α_j are year and industry (Fama–French 49) fixed effects. The variables are measured in the same year as RE status, because a loss or a payout in year t affects book equity in year t. Continuous variables are standardized each year to a mean of zero and a standard deviation of one, so that each coefficient measures the change in the probability of being an RE firm, in percentage points, associated with a one-standard-deviation change. Standard errors are clustered by firm.

Before reporting the estimated coefficients, we provide variable averages across RE and non-RE firms in Table 3. RE firms are much more often loss-making (76% vs. 31%) and less profitable, and their intangible investment is larger relative to total assets (22% vs. 12%). They repurchase slightly more relative to assets on average (1.4% vs. 1.2%) but pay less in dividends (0.5% vs. 1.0%), so that their total payouts relative to assets are about the same (2.1% vs. 2.2%). Their write-downs relative to assets are three times as large (2.4% vs. 0.8%), and they are smaller, more levered (0.61 vs. 0.22), and have a higher Tobin's Q (2.7 vs. 1.7).

Table 4 reports the estimates. Columns 1–3 add the loss, payout, and write-down variables in turn, Column 4 adds the controls, and Columns 5 and 6 add year and then industry fixed effects. The signs are stable across specifications: losses, intangible investment, the scale of payouts, and write-downs are positively related to RE status, and profitability neg-

atively. In the most complete specification (Column 6), a one-standard-deviation increase in intangible investment relative to assets raises the probability of being an RE firm by 3.2 percentage points, large relative to an unconditional RE frequency of about 5%. Payouts (1.2 percentage points) and write-downs (0.6) also predict RE status, while profitability lowers it (by 1.6 percentage points).[12] Once year and industry fixed effects are included, the loss indicator itself becomes small (0.5 percentage points), so that losses matter mainly through profitability. Intangible investment predicts RE status beyond current profitability partly because it is persistent and therefore captures the accumulated losses of firms that have expensed such investment for years.[13] The regressions and the counterfactuals of Figure 12 both rank intangible investment first and write-downs last, but losses rank higher in the regressions, where profitability has the second-largest effect in absolute value. The two approaches measure different things. In the counterfactuals, capitalizing intangible investment turns most deficits positive, and losses account only for the deficit that remains after all adjustments; in the regressions, profitability captures accumulated losses of all kinds, and its coefficient falls to zero once cumulative losses are controlled for. Logit regressions, reported as average marginal effects in Appendix Table A2, give the same signs and lead to the same conclusion: in Column 6, intangible investment remains the strongest positive predictor among the channel variables (2.1 percentage points), followed by the loss indicator (1.7) and payouts (0.7). The logit marginal effects are smaller than the linear probability estimates for most variables, and two differences stand out: the loss indicator is much larger (1.7 against 0.5 percentage points), and the effect of profitability is smaller (-0.5), so that in the logit, losses matter through the loss indicator as well as through profitability.

Overall, losses, large payouts, write-downs, and intangible investment all contribute to

[12] Entered separately, dividends and repurchases over assets are both significant in Column 6 (1.3 and 1.1 percentage points), and the difference between them is not statistically significant ($p = 0.27$).

[13] Adding cumulative losses relative to total assets to the specification of Column 6 lowers the coefficient on intangible investment from 3.2 to 2.3 percentage points (2.1 when cumulative payouts are added as well), while the coefficient on profitability falls from -1.6 to 0.0. Cumulative losses and payouts are measured over each firm's history in our sample.

the likelihood of negative book equity, and the expensing of intangible investment is the most important of them.

4.3 Changing characteristics or changing propensity?

The regressions describe which firms have negative equity, but not why their number has risen. The rise could reflect either more firms with the characteristics associated with negative equity, or a change in how firms with given characteristics end up with negative equity, for example because lenders and investors have become more tolerant of it. Following the approach of Fama and French (2001), we estimate a logit model of RE status in a base period, apply the estimated coefficients to the characteristics of firms in later years, and compare the predicted share of RE firms with the actual share. The characteristics are size (the within-year percentile of market value), profitability, a loss indicator, intangible investment, payouts relative to assets, Tobin's Q, and age; we exclude leverage, which is mechanically related to negative equity.

With 1975–1994 as the base period, the predicted share of RE firms rises from 2.0% in 1975 to 8.6% in 2025, while the actual share rises from 1.2% to 10.4% in the sample of firms with all characteristics: changing characteristics account for 72% of the rise. Because book equity reflects a firm's entire history of earnings and distributions, we also add cumulative payouts and cumulative losses relative to assets. The predicted share then reaches 13.1% in 2025, above the actual share. This specification is, however, close to an accounting identity, since cumulative losses and payouts are components of book equity, so the version with current characteristics only is the more informative test. With 1995–2004 as the base period, which allows us to include write-downs, current characteristics account for 56% of the rise since 1995, and the unexplained part is concentrated in 2022–2025. The rise of RE firms therefore mainly reflects changes in firm fundamentals—more loss-making firms and larger payouts—rather than a greater tendency of firms with given fundamentals to end up with negative equity.

4.4 Two kinds of red equity

The counterfactuals in Figure 12 overlap: a firm can turn positive after capitalizing intangibles *and* after adding back its buybacks. More importantly, they explain how many firms have negative equity, not what kind of firm an RE firm is. The expensing of intangibles is the largest source of negative equity for RE firms of all kinds: capitalizing intangibles turns more than half of RE firm-years positive whatever the source of their deficit (68% of payout-driven, 57% of write-down-driven, and 72% of operating-loss-driven firm-years since 1975; Table 5, Column 14). The intangibles adjustment therefore cannot separate healthy from fragile RE firms. We classify RE firms along a second dimension, the source of their deficit, using lifetime earnings as described in Section 2: whether a firm has earned money over its recorded history and returned more than it earned, or has accumulated losses. The classification uses only reported data, so it does not depend on how intangible capital is measured.

Table 5 describes the three types, over 1975–2025 (Panel A) and since 2010 (Panel B), and Figure 13 plots their evolution. The classification separates two very different groups. Payout-driven firms account for 17% of RE firm-years but 82.6% of RE market value over 1975–2025, and for 24% and 86.7% since 2010 (Panel B, Columns 2 and 3). Since 2010, they are large (71% are above the median market value, Column 11), mature (25 years since their IPO on average, median 22, Column 12), and profitable (78% report positive net income, against 57% of non-RE firms, Column 13; over the full sample 69% against 66%). Their RE status is persistent: 84% are still RE firms one year later and 60% five years later (Columns 8 and 10). Figure 14 shows that since 2010 payout-driven firms have accounted for 75–91% of the market value of RE firms each year, and on average for 77% of their assets and 81% of their employment since 2019.

Payout-driven firms are not all intangible-intensive. Some own much of the real estate they operate: since 2010, land and buildings at historical cost amount to a median 13% of total assets among payout-driven RE firms, twice the 6% of non-RE firms. Under U.S.

GAAP, land and buildings are carried at historical cost, less depreciation for buildings, and cannot be revalued upward, so the book value of property acquired long ago can be far below its market value, while lenders can rely on the cash flows and the economic value of these assets. Such firms can therefore borrow heavily and return the proceeds to shareholders even as their book equity becomes negative. McDonald's, which owns about 56% of the land and 80% of the buildings of its restaurants in the markets it consolidates, reports land and buildings at cost equal to 88% of its total assets, and its cumulative share repurchases exceed its total assets; its book equity has been negative since 2016. Lowe's, AutoZone, O'Reilly Automotive, and HCA Healthcare, whose land and buildings at cost amount to between 43% and 52% of total assets, are similar cases. Other large payout-driven firms are asset-light: Hilton and Marriott own or lease less than 2% of the hotel rooms in their systems and franchise or manage the rest, and their property, plant, and equipment amounts to about a tenth of their assets, so that their negative equity reflects buybacks and brands that are not recorded on the balance sheet. In both cases, the book value of a firm's assets understates their economic value—for real estate because it is carried at cost, for brands and other internally developed intangibles because they are expensed.

Operating-loss-driven firms are the mirror image: since 2010 they account for 71% of RE firm-years but only 11% of RE market value; only 14% of them are profitable and 19% are larger than the median firm, they have been listed for 10 years on average, and they are concentrated in pharmaceuticals and software. Their losses partly reflect the expensing of intangible investment: for the median firm, cumulative after-tax intangible investment equals 51% of its lifetime loss. But for most of them, the losses exceed all the intangible investment they have expensed: even if this investment were added back, only 22% would have non-negative lifetime earnings. Capitalization turns their book equity positive mainly because they have raised large amounts of capital from shareholders, not because they have earned it. Write-down-driven firms are few (72 firms and 5% of RE firm-years since 2010); their write-downs are mostly goodwill impairments (78% of the dollar amount), so their negative equity

stems from acquisitions whose recorded goodwill was later impaired. Because this group is small, the regression analyses below combine it with the operating-loss-driven group into *loss-driven* RE firms.

The two dimensions capture different things. Table 6 crosses the two types of RE firms with the sign of their equity once intangibles are capitalized. Within each intangible group, the two types differ as sharply as in Table 5: since 2010, 19% of payout-driven but 80% of loss-driven RE firm-years whose equity turns positive with intangibles are distressed, and 10% and 73% of those whose equity remains negative (Panel B, Column 4). Within each type, by contrast, the sign of intangible-adjusted equity matters much less than the source of the deficit (19% against 10% for payout-driven and 80% against 74% for loss-driven firms), and the same holds for delisting and profitability (Columns 6 and 7). The firms whose equity remains negative even after capitalization are not the most fragile: they include the most profitable RE firms, payout-driven firms whose distributions exceed even the intangible capital they have built (87% are profitable, and they account for 27% of RE market value since 2010). Capitalizing intangibles thus determines how many firms have negative equity, while the source of the deficit determines what negative equity means.

In short, the sign of book equity groups together two very different kinds of firms. Payout-driven RE firms are few in number but hold most of the market value of RE firms; they are large, mature, and profitable, and their negative equity reflects distributions to shareholders. Loss-driven RE firms are numerous but small, young, and mostly unprofitable, and their negative equity reflects accumulated losses. The rest of the paper shows that this distinction matters for distress, leverage, and valuation.

5 Are red equity firms distressed?

A second reason for excluding RE firms, besides their former rarity, is the presumption that negative equity signals financial distress (Fama and French, 1992; Griffin and Lemmon, 2002;

Vassalou and Xing, 2004). Section 4.4 showed, however, that most of the market value of RE firms belongs to large, profitable, payout-driven firms. This section asks whether RE firms are in fact distressed, and whether the answer differs between the two kinds of RE firms. We use the three distress indicators and the composite measure described in Section 2, an intangible-adjusted version of the composite, and performance-related delistings within two years as a realized outcome. Table 7 compares RE firms as a group with other firms, and Table 5 compares the types of RE firms.

As a group, RE firms are often distressed. Over 1975–2025, 69% of RE firm-years are classified as distressed, against 15% of non-RE firm-years, and 23% are followed by a performance-related delisting within two years, against 4% (Table 7, Panel A, Columns 2 and 7). This average, however, masks large differences between the two kinds of RE firms. Since 2010, 79% of operating-loss-driven RE firm-years are distressed and 18% are followed by a delisting (Table 5, Panel B, Columns 4 and 7). Payout-driven RE firms, by contrast, are distressed less often than non-RE firms (16% vs. 19%), although they are delisted somewhat more often (3.7% vs. 2.3%). Over the full sample, payout-driven firms are more often distressed than non-RE firms (23% vs. 15%; Panel A, Column 4).

When payout-driven firms are classified as distressed, it is mostly because of their leverage and their negative retained earnings, not because of weak earnings. Over 1975–2025, they are flagged by the Altman Z-score twice as often as non-RE firms (46% vs. 22%) and by the default probability twice as often (24% vs. 11%), but they have EBITDA below interest expense less often (14% vs. 25%). The Z-score penalizes them partly by construction: retained earnings enter the Z-score, and they are reduced by the repurchases of firms that retire the shares they buy back, so that 73% of payout-driven firm-years have negative retained earnings.

Standard distress measures also penalize intangible-intensive firms, because the accounting-based indicators treat expensed intangible investment as a cost. Among firms in the top tercile of intangible investment relative to assets, the share classified as distressed falls from

22% to 11% when the measures are adjusted for expensed intangibles. The firms that remain flagged after the adjustment are more likely to fail: 31% of them are delisted within two years, against 23% of the firms flagged by the reported measures. The adjustment, however, removes correct signals as well as false alarms: it flags 50% of the top-tercile firms that are later delisted, against 73% for the reported measures, and it predicts delisting less accurately overall (area under the ROC curve of 0.71 against 0.78). The reported measures thus flag many intangible-intensive firms that are not later delisted, but expensed intangible investment is not a reliable cushion against failure either.[14] The adjustment does not change the comparison of RE types: since 2010, 67% of operating-loss-driven RE firm-years are distressed on the adjusted measure, against 13% of payout-driven RE firms and 12% of non-RE firms (Table 5, Panel B, Column 5).

A further comparison puts these numbers in perspective: zero equity is not a natural threshold for distress. Researchers routinely keep firms with thin but positive equity (below 10% of assets), yet since 2010 these firms are more often distressed than payout-driven RE firms (40% vs. 16%) and more often delisted (7.1% vs. 3.7%). Payout-driven RE firms are therefore less distressed than firms with thin positive equity, which are kept in empirical analyses.

Overall, negative book equity is a poor indicator of financial distress. It identifies distress well for loss-driven RE firms, most of which are distressed, but not for payout-driven RE firms, which hold most of the market value of RE firms: they are distressed less often than the average listed firm and less often than firms with thin positive equity, and when standard measures flag them, it is mostly because of their leverage and buyback-depleted retained earnings. Their negative equity reflects a financing choice—distributions funded largely with debt—rather than accumulated losses, and the next section examines how they finance these distributions.

[14] Firms are sorted each year into terciles of intangible investment relative to total assets; the figures are pooled over 1975–2025.

6 Leverage

Section 5 showed that payout-driven RE firms are not distressed as a group, although standard measures flag some of them because of their leverage. This section asks whether the leverage of RE firms follows the same determinants as that of other firms. The question matters for two reasons. Leverage regressions usually exclude RE firms, whose book leverage is mechanically high, so it is unknown whether standard capital-structure relations hold for them. And the two leading theories make different predictions about profitability: under the trade-off theory, more profitable firms have more debt capacity and borrow more, whereas under the pecking-order theory (Myers and Majluf, 1984), more profitable firms use retained earnings and borrow less. If payout-driven firms borrow against their earnings to fund distributions, their leverage should rise with profitability.

We estimate standard leverage regressions (Rajan and Zingales, 1995; Frank and Goyal, 2009) on the pooled sample of RE and non-RE firms, in which each determinant, lagged by one year and standardized each year, is interacted with the RE indicator:

$$Lev_{i,t} = \alpha_t + \alpha_j + \beta' X_{i,t-1} + \delta\, RE_{i,t} + \gamma' RE_{i,t} X_{i,t-1} + \varepsilon_{i,t}.$$

The dependent variable is leverage (total debt over total assets), multiplied by 100.[15] The coefficients β measure the effect of each determinant for non-RE firms, and the interactions γ measure how the effect differs for RE firms. Table 8 reports the estimates, without fixed effects (Column 1), with year fixed effects (Column 2), and with year and industry fixed effects (Column 3).

Conditional on the determinants, RE firms have 47 percentage points higher book leverage (Column 3). Size, tangibility, industry leverage, intangible investment, and age have the same sign for both groups, but three determinants change sign. Tobin's Q, negatively

[15] When leverage is the dependent variable, we winsorize it at the 1st and 99th percentiles separately for RE and non-RE firms. The 99th percentile of the pooled distribution of book leverage is 0.96, which would truncate about a fifth of RE firm-years but almost no other firm-years.

related to book leverage for non-RE firms (−0.70), has a positive interaction of 8.34; taxes, which lower book leverage for non-RE firms (−4.09), have an interaction of 7.34; and past stock returns, slightly positive for non-RE firms, are negatively related to the book leverage of RE firms (interaction −2.89). For RE firms as a group, the profitability interaction is positive (+1.16) but not significant.

Because the two kinds of RE firms finance themselves so differently, we next allow the interactions to differ by type. We estimate

$$Lev_{i,t} = \alpha_t + \alpha_j + \beta' X_{i,t-1} + \sum_{g \in \{P,L\}} \left(\delta_g D^g_{i,t} + \gamma'_g D^g_{i,t} X_{i,t-1} \right) + \varepsilon_{i,t},$$

where $D^P_{i,t}$ and $D^L_{i,t}$ indicate payout-driven and loss-driven RE firm-years. In Table 9, Column 1 reports β, the coefficients for non-RE firms, and Columns 2 and 3 report γ_P and γ_L, the differences for each type; the coefficient for payout-driven firms, for example, is $\beta + \gamma_P$.

The profitability result comes entirely from payout-driven firms. Their interaction with profitability is +14.93 (Column 2), against −1.09 for loss-driven firms (Column 3; the difference is significant at the 0.1% level). More profitable payout-driven firms therefore have higher book leverage, with a total effect of +14.62 percentage points per standard deviation of profitability, compared with −0.31 for non-RE firms (Column 1). Consistent with the trade-off theory, and contrary to the pecking-order pattern of other firms, their book leverage rises with profitability, which is what one would expect if they borrow against their earnings to fund distributions. Size also enters with opposite signs: larger payout-driven firms have relatively less book leverage (−10.62, Column 2), larger loss-driven firms relatively more (+5.66, Column 3).

In sum, the leverage of RE firms is not simply a mechanical consequence of negative equity. For payout-driven RE firms, book leverage rises with profitability, consistent with the trade-off theory and with borrowing against earnings to fund distributions, whereas other firms follow the pecking-order pattern. Leverage regressions that exclude RE firms therefore

miss a group of large firms whose financing policy differs systematically from that of the firms they keep.

7 Empirical implications

The previous sections showed that negative book equity is largely an accounting outcome and that it groups together two very different kinds of firms. This section draws out what this implies for accounting and for research that relies on book equity. We discuss the capitalization of internally generated intangibles (Section 7.1) and the firms that the symmetric expensing of acquired intangibles would add to the RE group (Section 7.2), examine whether book value remains value relevant when it is negative (Section 7.3), and show how the treatment of RE firms affects an intangible-adjusted value factor (Section 7.4).

7.1 Capitalizing internally generated intangibles

The asymmetric treatment of acquired and internally generated intangibles is the single largest source of negative book equity: capitalizing intangible investment would remove 71% of RE firm-years since 1995 (Section 4.2), and the result holds for all intangible capital measures we use. Two otherwise identical firms—one that acquires its intangibles, one that builds them—can thus fall on opposite sides of the zero line and be treated differently in empirical work. The FASB's agenda consultation on intangibles (Financial Accounting Standards Board, 2024) raises exactly the question of narrowing this gap (see Lev and Gu, 2016; Penman, 2024, for opposing views). Our evidence adds a caveat: capitalization would also turn most loss-driven RE firms positive, without changing their much higher distress and delisting rates. Capitalized book equity therefore measures the accounting understatement of the firm's capital, not its financial health.

7.2 Acquired intangibles and tangible equity

Section 3 showed that the share of RE firms would triple if acquired intangibles were expensed like internally generated ones (Figure 2). Which firms would this add to the RE group? Table 10 compares three groups: firms with positive tangible equity, firms with positive book equity but negative tangible equity, and RE firms. Since 2010, 18.7% of firm-years have positive book equity but negative tangible equity, and these firms account for 28.7% of market value (Panel B, Columns 2 and 3). They are acquirers: intangible assets on the balance sheet make up 50% of their total assets, against 12% for firms with positive tangible equity, and 61% of these intangibles are goodwill (Columns 4 and 5). They are more levered (book leverage of 0.40 against 0.17, Column 6), more often profitable (65% against 55%, Column 7), and larger (70% are above the median market value, against 47%, Column 8). Yet they are only slightly more often distressed (15% against 12% on the intangible-adjusted measure, Column 10), and they are delisted for performance reasons at the same rate (2.4% against 2.3%, Column 11). RE firms, by contrast, are distressed in 53% and delisted in 14% of firm-years. Negative tangible equity, in other words, mostly identifies large, profitable firms that have grown through acquisitions. Prominent examples in 2025 include Broadcom, Oracle, Johnson & Johnson, Procter & Gamble, IBM, Cisco, Thermo Fisher Scientific, and Pfizer, all profitable. Broadcom, for instance, reports book equity of $81 billion, but intangible assets, three quarters of them goodwill (mostly from its acquisition of VMware), make up 76% of its total assets; without them, its equity would be −$49 billion. IBM's book equity of $33 billion would similarly become −$46 billion.

These results complete the picture of the capitalization debate. Expensing acquired intangibles to make the treatment symmetric would triple the share of firms with negative equity and add mostly large, healthy acquirers to it; capitalizing internally generated intangibles would shrink it to a few percent. Under either symmetric treatment the sign of book equity would mean something different from what it means today, which reinforces our conclusion that the source of negative equity, not its sign, is informative.

7.3 The value relevance of negative book value

In the valuation framework of Ohlson (1995), a firm's market value depends on its book value and its earnings, and a large accounting literature examines how each is priced. Book value matters most when earnings are uninformative, as for loss firms and firms in poor financial health, because it measures the resources that could be abandoned or redeployed (Burgstahler and Dichev, 1997; Barth et al., 1998; Collins et al., 1999b), and the relative importance of book value and earnings has changed over time (Collins et al., 1997; Barth et al., 2023). Negative book equity challenges this role. For loss-driven RE firms, book value still reflects the resources that remain after accumulated losses. For payout-driven firms, however, a more negative book value reflects more cash returned to shareholders by a profitable business, not fewer resources, so book value need not be positively priced. For both types, moreover, reported book value omits the intangible capital that the firm has built, which is the main source of negative equity (Section 4).[16]

Table 11 regresses the share price on book value and earnings per share, separately for positive-equity firms (Column 1), payout-driven RE firms (Column 2), and loss-driven RE firms (Column 3). Panel A uses reported book value and earnings. Panel B adds intangible capital to book value and adjusts earnings accordingly, adding back intangible investment net of the amortization of intangible capital and of taxes at the firm's effective tax rate.[17] Both panels use the same firm-years, and the last two rows of each panel test, in a pooled regression in which book value and earnings are interacted with indicators for the two types of RE firms and the year and industry fixed effects are specific to each group, whether the book-value coefficient of payout-driven firms differs from that of positive-equity firms and

[16] Jan and Ou (2012) show that RE firms are valued more highly than their book values suggest, but they do not distinguish the sources of the deficit.

[17] Intangible capital is added to book value at its pre-tax amount, as elsewhere in the paper, which ignores the deferred taxes that capitalizing intangibles for financial reporting, but not for tax purposes, would create. When it is added net of taxes at the firm's effective tax rate, the coefficient on adjusted book value per share is 0.64 for positive-equity firms, 0.68 for payout-driven firms, and 0.24 for loss-driven firms, all significant at the 1% level, and the coefficient of payout-driven firms still does not differ from that of positive-equity firms ($p = 0.68$).

from that of loss-driven firms.

Reported book value is positively priced for positive-equity firms (1.14 in Panel A) but negatively priced for RE firms of both types. The negative relation is strongest for payout-driven firms: their coefficient is −3.59, against −1.83 for loss-driven firms, and the difference is significant at the 5% level. Earnings carry the valuation of payout-driven firms, with a coefficient of 7.59, against 4.90 for positive-equity firms; the earnings of loss-driven firms, by contrast, are barely priced (0.98), the pattern of loss firms whose losses reflect expensed investment (Collins et al., 1999b; Gu et al., 2023).

Recognizing intangible capital restores the usual relation (Panel B). Intangible-adjusted book value is positively priced in all three groups: the coefficient is 0.51 for positive-equity firms, 0.62 for payout-driven firms, and 0.24 for loss-driven firms. For payout-driven firms, we cannot reject that it equals that of positive-equity firms ($p = 0.28$): once the intangible capital they have built is recognized, their book value is priced like that of other firms. For loss-driven firms, adjusted book value is also positively priced, but at less than half the rate of the other two groups (the difference from payout-driven firms is significant at the 1% level). The results are similar when all variables are scaled by total assets instead of per share (Appendix Table A3): the coefficient on reported book value is 1.80, −2.11, and −0.43 for the three groups, and that on adjusted book value 0.54, 0.74, and 0.34. They are also similar with the other measures of intangible capital.[18]

In short, reported book value is negatively priced for RE firms, most strongly for payout-driven firms, whose market value rests on their earnings. Once the intangible capital that these firms have built is recognized, book value is positively priced again, and for payout-driven firms at a rate statistically indistinguishable from that of positive-equity firms. For RE firms, intangible-adjusted book equity is thus not only positive more often than reported

[18] With the measures of Ewens et al. (2025), Eisfeldt and Papanikolaou (2013), and Iqbal et al. (2025), intangible-adjusted book value per share is positively priced in all three groups, with coefficients of 0.78, 0.27, and 0.63 for payout-driven firms and 0.30, 0.12, and 0.20 for loss-driven firms. When the variables are scaled by total assets, the coefficient for payout-driven firms is not significant with the measure of Iqbal et al. (2025) (0.10).

equity, but also value relevant.

7.4 Red equity and the intangible-adjusted value premium

Asset-pricing studies routinely drop RE firms when they form value portfolios, and studies of intangible-adjusted value factors either exclude them explicitly (Park, 2022; Iqbal et al., 2025) or do not state how they are treated (Eisfeldt et al., 2022; Gulen et al., 2025). An exception is Stambaugh and Yuan (2017), whose mispricing factors are built from a stock universe that includes RE firms (Soebhag et al., 2024), with breakpoints computed from all NYSE, AMEX, and NASDAQ stocks. We focus on intangible-adjusted value factors for two reasons. First, the expensing of internally generated intangibles is the main source of negative book equity (Section 4): for most RE firms, reported book equity understates the capital they have invested, and their reported book-to-market ratio cannot be interpreted as usual. Adding intangible capital to book equity addresses exactly this problem, so value factors built on intangible-adjusted book equity are the natural setting in which to ask how RE firms should be treated. Second, intangible-adjusted equity separates RE firms into two groups. About 70% of RE firm-years have positive equity once intangible capital is added (Table 5, Column 14) and can be sorted on a meaningful book-to-market ratio; the others have negative equity even after the adjustment. This allows us to add RE firms in two steps and to see which of them matter.

We add the intangible capital of Peters and Taylor (2017) to book equity and form six value-weighted portfolios each June from two size groups (split at the median market value) and three groups of book-to-market (split at the 30th and 70th percentiles), with breakpoints computed from all stocks;[19] the value factor is the average return on the two value portfolios minus the average return on the two growth portfolios. The timing follows Fama and French

[19]Breakpoints computed from all stocks are the most common choice in the literature: of the 109 published studies whose anomalies Hou et al. (2020) replicate, 68% compute breakpoints from all stocks, 13% from NYSE stocks only, and 19% report both (Walter et al., 2024). With NYSE breakpoints, the intangible-adjusted factor itself earns no FF alpha (-0.02, $t = -0.02$), and including RE firms makes no significant difference.

(1993) and ensures that the portfolios use only information available when they are formed. Accounting data are assumed to become public six months after the fiscal year end, so that the portfolios formed at the end of June of year t use book equity and intangible capital from the most recent fiscal year that ended at least six months earlier, that is, no later than December of year $t-1$. Book-to-market divides this book equity by the firm's market value of equity (summed over its share classes) at the end of June of year t, the date at which size is also measured; unlike Fama and French (1993), who use market equity at the end of December of year $t-1$, we thus use market equity at the formation date, which is equally known when the portfolios are formed. The portfolios are held from July of year t to June of year $t+1$, and returns are value-weighted by the market value at the end of the previous month. The intangible capital stocks are accumulated from each firm's current and past accounting data; apart from the interpolation of missing values and the backcasting of early R&D described in Appendix A, they use no information from later years. The exception is the measure of Iqbal et al. (2025), whose estimation uses future sales (Appendix A) and which we use only as a robustness check. Table 12 compares four strategies that differ only in the book equity used and the firms included: the standard HML (Column 1), and three intangible-adjusted versions that exclude firms with negative intangible-adjusted equity (HML_INT, Column 2), exclude firms with negative reported equity (HML_INT ex RE, Column 3), or include all firms (HML_INT inc RE, Column 4). Column 5 reports the difference between Columns 4 and 3, the effect of including RE firms.

Including RE firms raises the intangible-adjusted value premium by 0.47 percentage points per year over 1970–2026 ($t = 2.55$), its CAPM alpha by 0.57 points ($t = 3.24$), and its alpha with respect to the market, size, profitability, and investment factors of Fama and French (2015) (FF alpha) by 0.75 points ($t = 3.89$; Panel A, Column 5). These effects are statistically significant, and they are growing. Since July 2009, including RE firms raises the premium by 0.82 points ($t = 1.62$), the CAPM alpha by 1.18 points ($t = 2.45$), and the FF alpha by 1.30 points ($t = 2.63$; Panel B, Column 5), almost twice the full-sample

effects.[20] To check whether the effect accumulates steadily or comes from a few unusual months, Figure 15 plots the cumulative sum of the monthly difference between HML_INT inc RE and HML_INT ex RE, together with its cumulative CAPM and FF abnormal returns, that is, the monthly difference minus its exposure to the factors. The abnormal returns accumulate steadily over the whole sample rather than in a few months, although the largest single month, June 2025, adds 3.8 percentage points. The FF abnormal return lies above the CAPM abnormal return, and both above the raw difference, because including RE firms slightly lowers the factor's loadings on the market, size, profitability, and investment factors (by between 0.013 and 0.024), all of which earned positive premia over the sample; risk adjustment therefore raises rather than lowers the effect. This is not the result of a particular cut-off: Figure 16 shows that the effect is generally larger in samples that start later. The FF alpha difference rises from 1.1 percentage points for samples starting in 2000 to 1.7 points for samples starting in 2018 (the CAPM alpha difference from 0.9 to 1.6 points), and it is significant at the 5% level for all start years except 2015 and 2017, for which it is significant at the 10% level. The magnitude of the effect thus keeps rising as the sample starts later, while its statistical significance declines somewhat. The lower significance is to be expected: later start years leave far fewer monthly observations, so the estimates become less precise and the confidence interval widens, even though the estimated effect is larger. The pattern holds for all intangible capital measures (Table A4): the difference is positive for every measure, its alphas are significant for the three measures available from 1970, and it is larger since 2009 for all of them.[21] Both types of RE firms contribute: including only payout-driven or only

[20] The results are robust to adding the momentum factor of Carhart (1997) to the factor regression. With momentum, the alpha of the difference between HML_INT inc RE and HML_INT ex RE is 0.61 percentage points per year over the full sample ($t = 3.05$), against 0.75 without momentum ($t = 3.89$), and 1.18 points since July 2009 ($t = 2.30$), against 1.30 ($t = 2.63$). Appendix Figure A4 repeats Figure 16 with momentum included.

[21] The measure of Ewens et al. (2025) is the only one for which including RE firms yields no significant alpha: the differences are positive but insignificant in both periods (FF alpha 0.40, $t = 1.48$, over 1976–2026, and 0.73, $t = 1.32$, since July 2009). For the measure of Iqbal et al. (2025), the FF alpha is significant over 1980–2026 (0.71, $t = 2.03$) and the CAPM alpha since July 2009 (1.27, $t = 1.97$), while the CAPM alpha over the full sample is not ($t = 1.28$).

loss-driven RE firms raises the FF alpha by 0.24 percentage points ($t = 2.70$) and 0.47 points ($t = 2.75$), less in each case than including both (Table 13, Panel A, Columns 6 and 7). The effect therefore does not come only from loss-driven RE firms. This is important because, unlike loss-driven RE firms, payout-driven RE firms are profitable and mostly large, and their negative equity reflects distributions to shareholders rather than losses (Section 4.4).

The two steps matter at different times. Adding the RE firms whose equity is positive after the adjustment (HML_INT relative to HML_INT ex RE) raises the CAPM alpha by 0.33 percentage points over the full sample ($t = 2.66$), but not significantly since 2009 (0.40, $t = 1.23$). Adding the firms whose equity remains negative after the adjustment (HML_INT inc RE relative to HML_INT) raises the CAPM alpha by 0.24 points over the full sample ($t = 1.69$; FF alpha 0.42, $t = 2.69$) and by 0.78 points since 2009 ($t = 2.02$). The recent increase in the effect therefore comes mostly from the firms whose deficit remains even when intangible capital is counted as an asset. Because their adjusted book-to-market is the lowest, these firms enter the growth portfolios and lower the breakpoints, so that their effect operates through the composition of the portfolios rather than through their ranking.

Where does the effect come from? Including RE firms changes the portfolios in three ways: the RE firms themselves enter them; because most RE firms have the lowest book-to-market, the 30th and 70th percentile breakpoints fall, so that the growth firms closest to value move to the neutral portfolios; and some neutral firms move into the value portfolios. We decompose the change in the return of each leg exactly into the contributions of these groups.[22] For the intangible-adjusted factor, the growth firms that leave the growth portfolios earn higher returns than those that remain (14.9% against 7.7% per year on average), so their departure raises the factor by 0.30 percentage points per year. The RE firms that enter

[22] Within each month and size group, the value-weighted return of a leg that includes RE firms equals the return of the leg that excludes them plus, for each group of firms that enters the leg, its weight in the leg times the difference between its return and that of the firms that are in the leg in both versions, minus the corresponding term for each group that leaves; as in HML, the legs average the two size groups. The decomposition uses the same portfolios and months as Table 12, and the contributions add up to the change in the factor return.

the growth portfolios change it little (by 0.03 points). At the value end, about a fifth of RE firm-years have an intangible-adjusted book-to-market high enough to be placed in the value portfolios; these firms earn high returns (23.8% per year on average, against 16.9% for the other value firms) and raise the factor by 0.09 points, and the non-RE firms that enter the value portfolios add 0.04 points. In total, the return of the growth leg falls by 0.33 percentage points per year (CAPM alpha -0.43, $t = -2.50$) and that of the value leg rises by 0.14. Since July 2009, the growth leg falls by 0.51 points, 0.39 from the firms that leave it and 0.12 from the RE firms that enter it, and the value leg rises by 0.31.

The same decomposition explains why including RE firms has no comparable effect on the standard HML, which is built on reported book equity. With the same construction, including RE firms changes the standard HML by -0.10 percentage points per year over 1970–2026 ($t = -0.45$), its CAPM alpha by 0.02 points ($t = 0.12$), and its FF alpha by 0.22 points ($t = 1.08$); since July 2009, none of these differences is significant either. The reclassification at the growth end is the same as in the intangible-adjusted factor: the growth firms that leave the growth portfolios raise the standard HML by 0.30 points per year. Two other effects, however, offset it. First, on reported book-to-market, every RE firm is placed in the growth portfolios, including many intangible-intensive firms whose reported book equity understates the capital they have invested: the median intangible capital of RE firms in the growth portfolios equals 62% of their market value, against 14% for the other growth firms. These firms earn higher returns than the other growth firms, so that adding them lowers the factor by 0.13 points. Second, the firms that enter the value portfolios earn lower returns than the other value firms (13.8% against 15.8% per year on average), which lowers the factor by 0.26 points; in the intangible-adjusted factor, the firms that enter the value portfolios, among them RE firms with large intangible capital, instead earn higher returns. The difference between the two factors thus comes from where RE firms are placed: reported book equity puts all of them at the growth extreme, whereas intangible-adjusted book equity moves about a third of them out of the growth portfolios, a fifth into the value portfolios,

and leaves in the growth portfolios RE firms whose returns are similar to those of other growth firms.

These effects are measured over a sample in which RE firms made up fewer than 9% of firms in every year before 2022. Their share has risen sharply only in recent years, and the payouts and intangible investment that drive negative equity continue to grow. The consequences of excluding RE firms for empirical analyses, and for the inferences drawn from them, are therefore likely to grow if these trends continue.

7.5 Implications for empirical research

Because the two kinds of RE firms differ so much, the sign of book equity is a poor screen. Excluding RE firms removed $2.6 trillion of market value from the sample in 2025, most of it in profitable and mature payout-driven firms, which account for 82.5% of the market value of RE firms over 1975–2025 and 86.5% since 2010; it also treats a financing choice as a sign of distress. For RE firms, book-based ratios such as the book-to-market ratio or the return on equity lose their usual interpretation, so studies that rely on these ratios should either use intangible-adjusted equity, whose book value is positively priced for RE firms of both types (Section 7.3), or report results with and without RE firms. Studies of distress should rely on direct measures of distress rather than on the sign of equity, and the source of the deficit, measured with lifetime earnings, provides additional information about a firm's condition. Finally, studies that construct value factors from intangible-adjusted book equity should report how RE firms are treated, because including them raises the premium and its alphas significantly, and increasingly so.

8 International evidence

The previous sections documented the rise of RE firms in the United States and traced it largely to the accounting treatment of intangibles and to large payouts. This section asks

whether the same holds in other countries. The comparison is informative because other countries differ in accounting standards, in payout practices, and in listing rules, so it shows which of our findings are specific to the United States.

Several institutional differences lead us to expect fewer RE firms outside the United States. First, IFRS, which most listed firms in Europe, Australia, and Korea use, requires development costs that meet certain criteria to be capitalized (IAS 38), whereas U.S. GAAP expenses almost all research and development. Second, IFRS allows property, plant, and equipment to be carried at revalued amounts (IAS 16) and investment property at fair value (IAS 40), whereas U.S. GAAP requires historical cost, so that the undervaluation of real estate that allows some U.S. firms to borrow and pay out beyond their book equity (Section 4.4) can be smaller. Third, IFRS requires impairment losses on assets other than goodwill to be reversed when the asset's recoverable amount increases (IAS 36), whereas U.S. GAAP prohibits such reversals for assets held and used. Fourth, and outside accounting, company law in the European Union prohibits distributions that would reduce net assets below subscribed capital plus undistributable reserves (Directive (EU) 2017/1132, Article 56), and public companies in the United Kingdom are subject to the same restriction (Companies Act 2006, section 831). These rules constrain payout-driven deficits, although they apply to each company's own annual accounts rather than to a group's consolidated accounts. All four differences work against negative equity abroad. Finding RE firms at rates similar to those in the United States would therefore be consistent with the asymmetric treatment of intangibles, which IFRS shares for all internally developed intangibles other than development costs, remaining important despite them.

We use Compustat Global for listed firms outside the United States over fiscal years 1995–2025, with the same industry exclusions as in our main sample (utilities, banks, and insurance companies). Three differences matter for the comparison. First, common equity is reported under each firm's accounting standards: most listed firms in Europe, Australia, and Korea report under IFRS, and most firms in Japan and China use domestic standards.

Second, Compustat Global does not contain market prices, so we measure size by total assets. Third, coverage varies across countries and over time. We assign firms to countries by their country of incorporation and define Europe as the 27 EU member states, the United Kingdom, Switzerland, Norway, and Iceland; the European figures pool all listed firms in these countries, so that each country is weighted by its number of listed firms. Table 14 reports Europe as a whole, every European country with at least 1,000 listed firm-years over 2015–2025, and the five largest other markets; Table A5 in the appendix reports the remaining non-European countries with at least 1,000 listed firm-years. We exclude India, whose RE share jumps from 0.6% in 2010 to 7.1% in 2011 because most Indian firms with negative equity in 2011 had positive equity a year earlier. The jump coincides with India's Revised Schedule VI, which from fiscal 2011–12 required accumulated losses that had previously been shown on the assets side of the balance sheet to be deducted from shareholders' funds—another example of how reporting rules, rather than firms, can change how many firms have negative equity. We also exclude firms incorporated in Bermuda, the Cayman Islands, and the British Virgin Islands.

Negative book equity is not only a U.S. phenomenon. Figure 17 plots the share of RE firms since 1995 for the United States, Europe as a whole, and the countries covered continuously since 1995. In Europe as a whole, the share of RE firms rose from 2–3% in the late 1990s to between 5.5% and 7.1% since 2019. In the United Kingdom, France, and Australia it is close to the U.S. level (9.4%, 10.6%, and 9.1% in 2025, against 10.5%; Table 14, Column 3), while it is lower in Germany (2.0%). By contrast, negative book equity is almost absent in Japan, Korea, China, and Taiwan, where at most 1.1% of listed firms had negative equity in 2025. In Japan, exchange rules delist firms whose liabilities exceed their assets for an extended period, which also limits the number of listed firms with negative equity. Excluding the smallest fifth of firms by assets lowers these shares in most countries, but RE firms remain common in the United Kingdom, France, and Australia and rare in East Asia (Columns 4 and 5).

Outside the United States, negative equity results much less often from payouts. Over 2015–2025, 11% of European RE firm-years are payout-driven, against 24% in the United States, and the share is at most 6% in Australia, Japan, Korea, China, and Taiwan (Column 8). Negative book equity abroad thus reflects accumulated losses even more than in the United States.

The accounting result of Section 3 holds internationally. If all intangible assets on the balance sheet, including goodwill, were expensed, the share of firms with negative equity in Europe would rise from 6.2% to 23.5% in 2025, and to 28.9% in the United Kingdom, 32.3% in France, 27.9% in Sweden, and 29.8% in the Netherlands—close to the 30.6% of the United States (Column 7). In East Asia, the share would remain at about 3% or less. In Europe as in the United States, the intangible assets recognized on the balance sheet keep a large group of firms on the positive side of the zero line.

Overall, the international evidence supports our main conclusions. Negative book equity is about as common in the United Kingdom, France, and Australia as in the United States, and in Europe, as in the United States, expensing all intangible assets recognized on the balance sheet would raise the share of firms with negative equity several-fold. Where negative equity is rare, as in East Asia, it remains rare under this treatment as well. The large group of profitable, payout-driven RE firms, by contrast, is mainly a U.S. phenomenon: abroad, negative equity reflects accumulated losses far more often than distributions to shareholders.

9 Conclusion

Negative book equity has moved from a curiosity to a common feature of U.S. listed firms: more than one in ten listed firms has negative book equity, and these firms were worth $2.6 trillion in 2025. How many firms have negative equity is largely an accounting outcome. GAAP capitalizes intangibles that firms buy but expenses those they build, and applying either rule consistently would leave between 2.5% and 31% of listed firms with negative book

equity. Under current rules, the rise reflects the expensing of internally generated intangibles, losses, large-scale share repurchases, and write-downs.

Behind the sign of book equity are two distinct groups of firms. Payout-driven firms are few in number but account for most of the value. They are mature and profitable, their negative equity reflects distributions in excess of lifetime earnings financed largely with debt, and their reported book value is negatively priced, whereas their intangible-adjusted book value is priced like that of other firms. Firms whose deficit comes from operating losses are far more numerous, but they are small, young, and fragile. Treating negative book equity as a sign of distress, or excluding RE firms altogether, discards this distinction, and excluding them already changes intangible-adjusted value premia by amounts that have roughly doubled since 2009. The phenomenon is not specific to the United States: negative equity is almost as common in the United Kingdom, France, and Australia, and in Europe, as in the United States, expensing the intangible assets on firms' balance sheets would multiply the share of firms with negative equity.

More broadly, whatever their effect on inference in historical samples, RE firms now make up a sizable and economically important part of the stock market, and they are likely to become more common as intangible investment and share repurchases continue to grow. Historical analyses are therefore likely to understate the consequences of excluding them, and these consequences will grow as the same forces continue to shape firms' balance sheets. We hope our evidence encourages researchers to treat RE firms as a meaningful and growing segment of the market rather than as outliers to be discarded, and to look beyond the sign of book equity to its source.

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Figure 1: Proportion of red equity firms

This figure plots the share of firms with negative book equity among U.S. listed firms (excluding utilities, banks, and insurance companies).

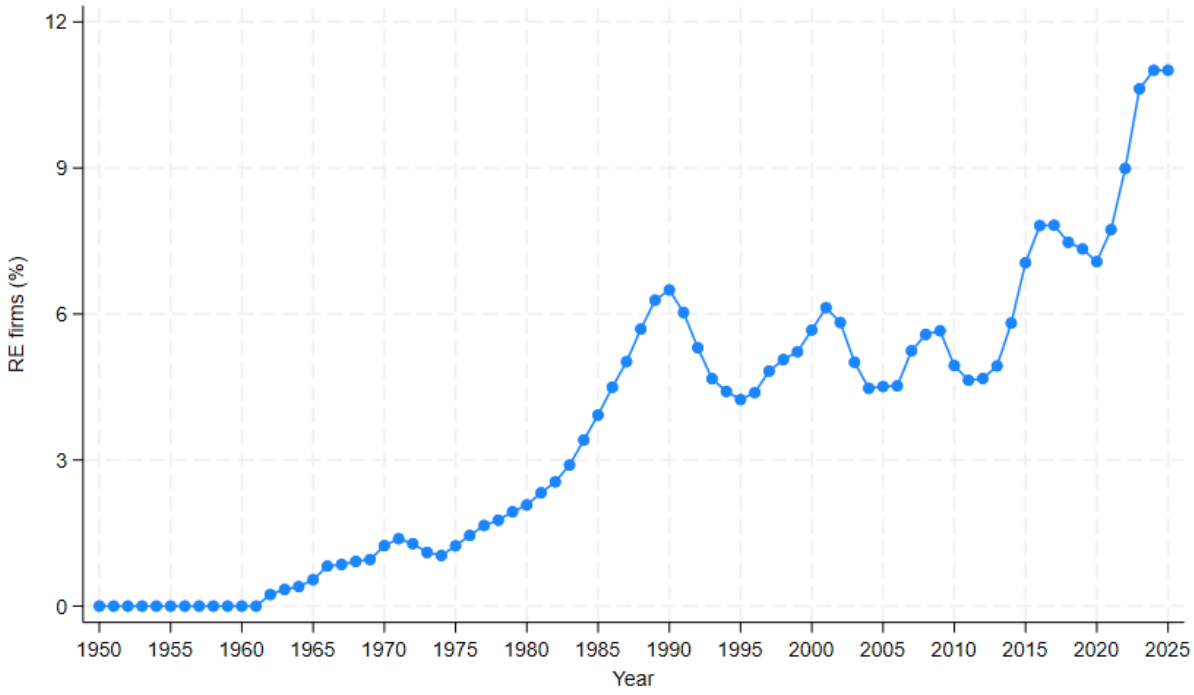


Figure 2: Red equity under three accounting treatments of intangibles

This figure plots the share of firms (panel a) and of total market capitalization (panel b) with negative equity under three treatments of intangibles. All intangibles capitalized: common equity plus internally generated intangible capital is negative. Current GAAP: common equity is negative. No intangibles capitalized: tangible equity, common equity minus intangible assets on the balance sheet (including goodwill), is negative. The three series use the same firm-years.

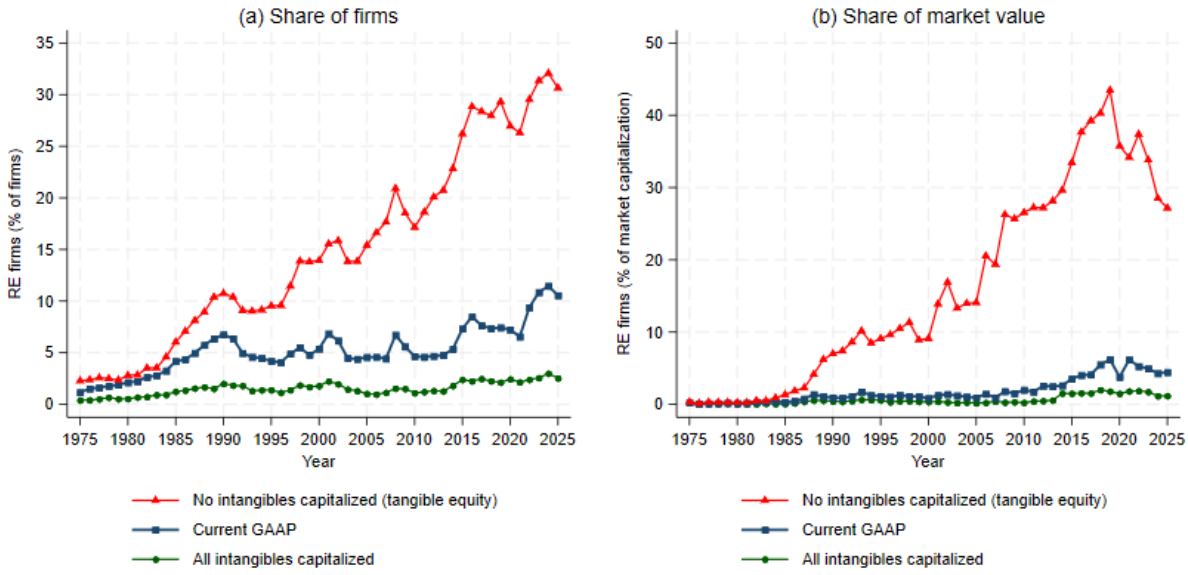


Figure 3: The magnitude of red equity

This figure plots the yearly 25th, 50th, and 75th percentiles of the equity-to-assets ratio of red equity firms (three-year moving averages).

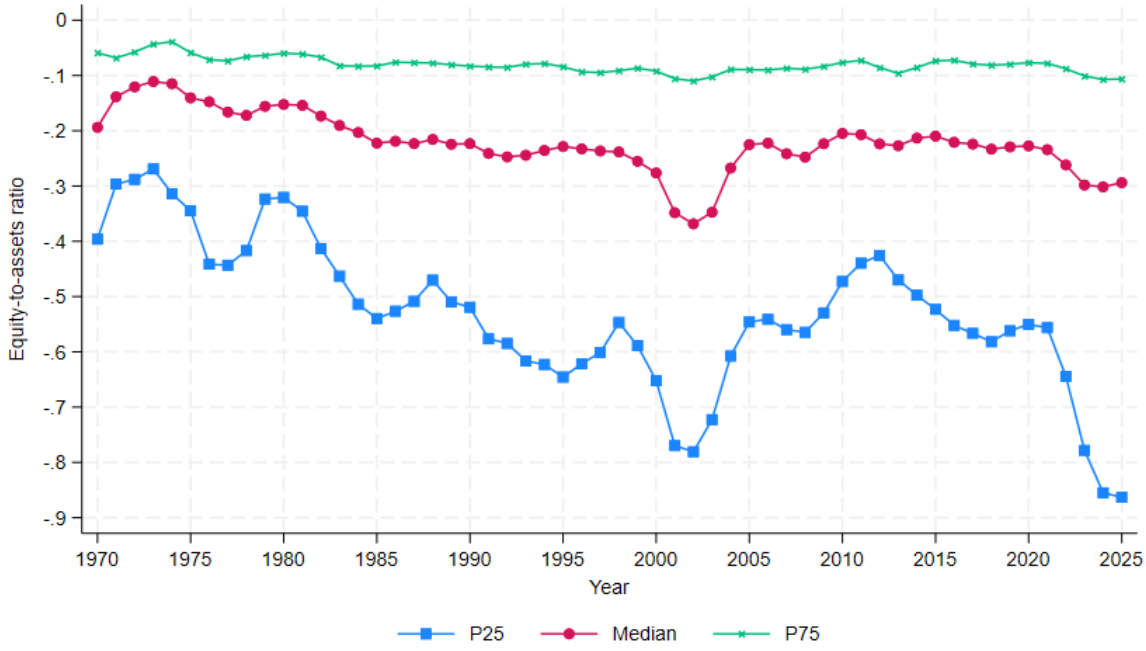


Figure 4: Red equity firms by size

This figure plots the share of small and large red equity firms; small (large) firms have a market capitalization below (above) the yearly median.

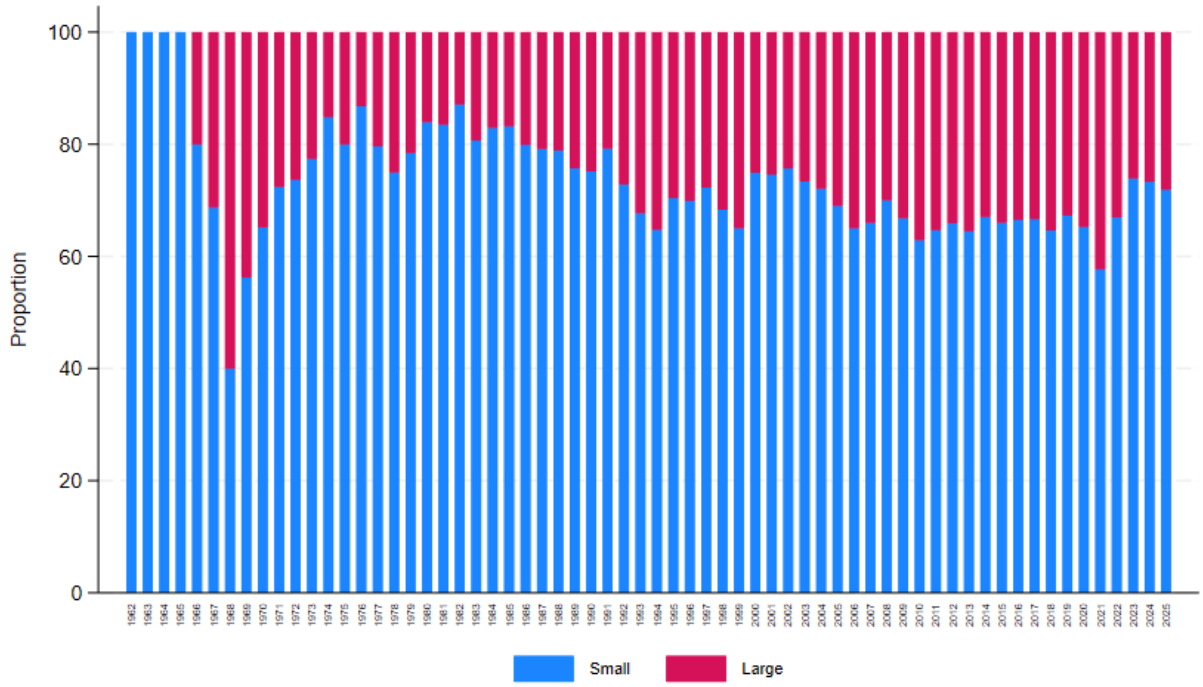


Figure 5: Red equity firms by age

This figure plots the share of red equity firms by years since IPO.

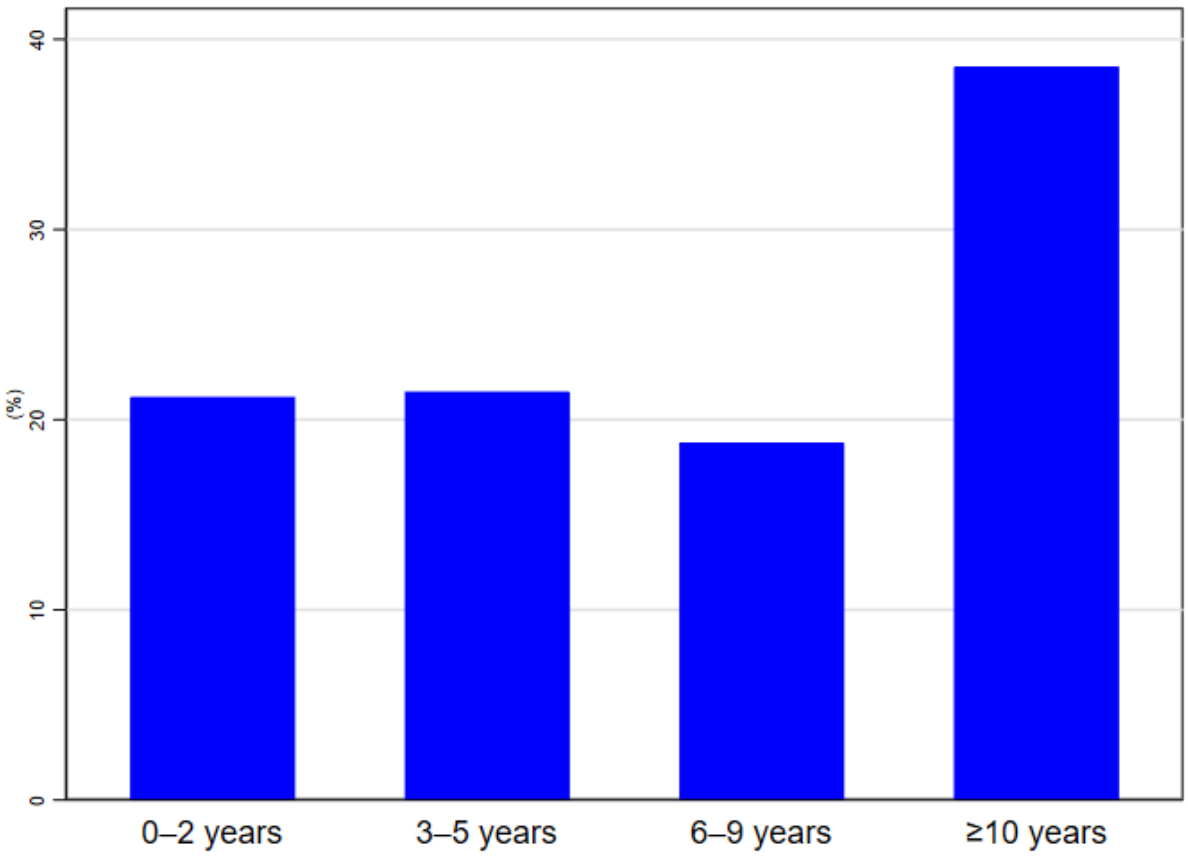
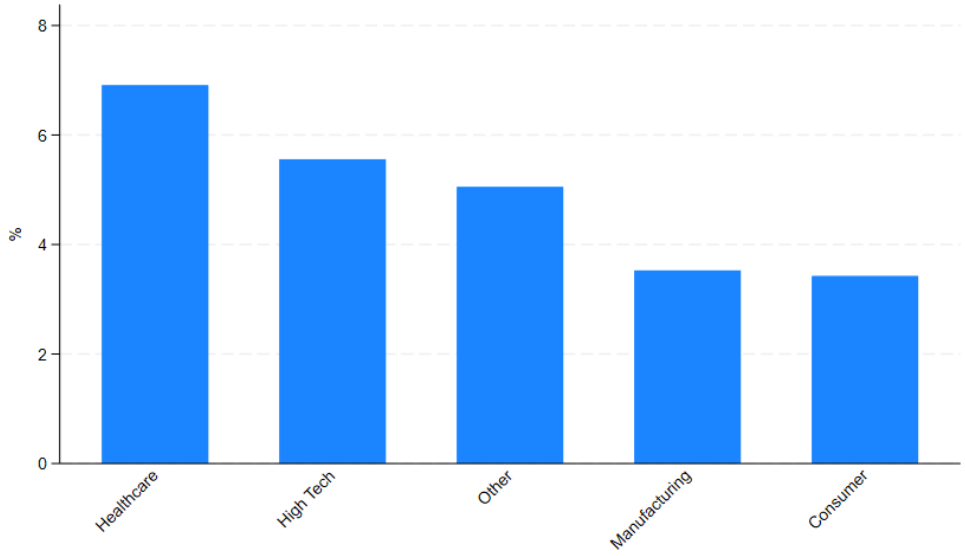


Figure 6: Red equity firms by industry

This figure shows the proportion of red equity firms by Fama–French five-industry classification, (a) over all years and (b) in 2025.

(a) All years



(b) 2025

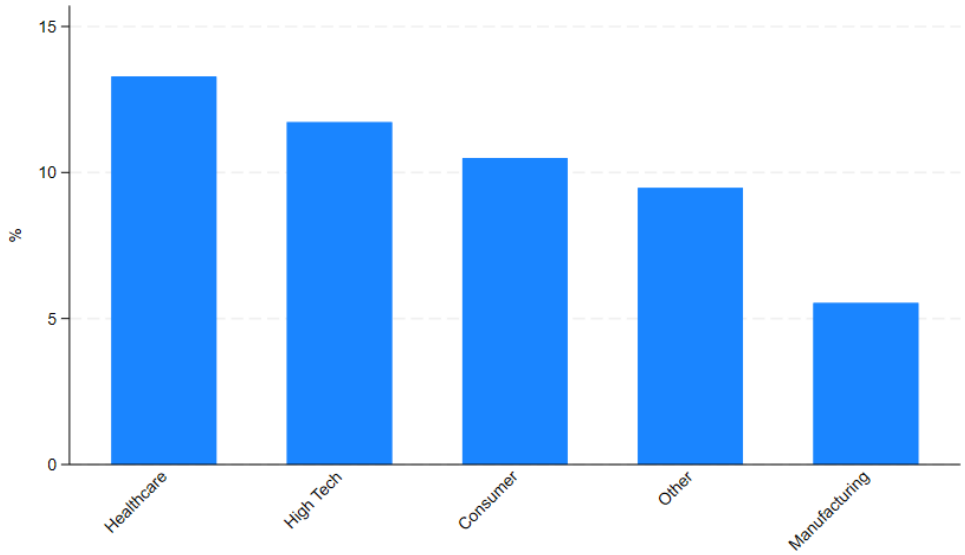


Figure 7: Proportion of money-losing firms

This figure plots the yearly share of firms with negative net income (three-year moving average).

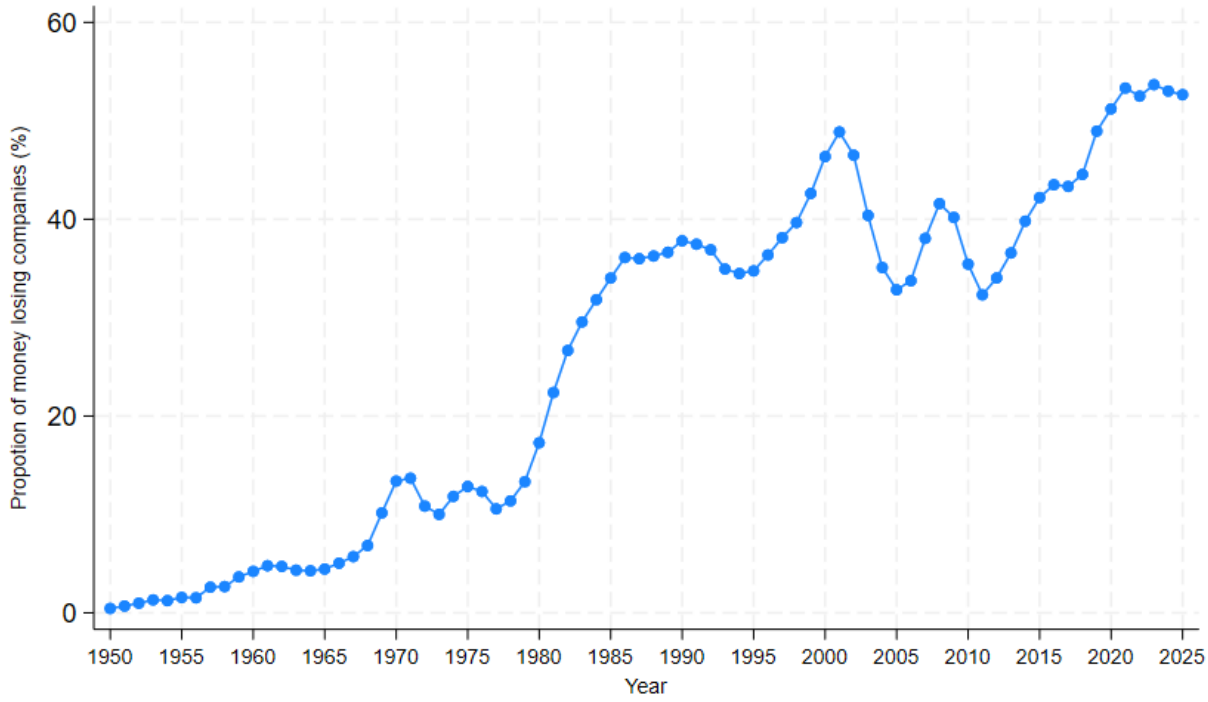


Figure 8: Money-losing firms with and without the expensing of intangible investment

This figure plots the share of firms with negative GAAP net income and with negative net income after adding back after-tax intangible investment (R&D plus 30% of SG&A).

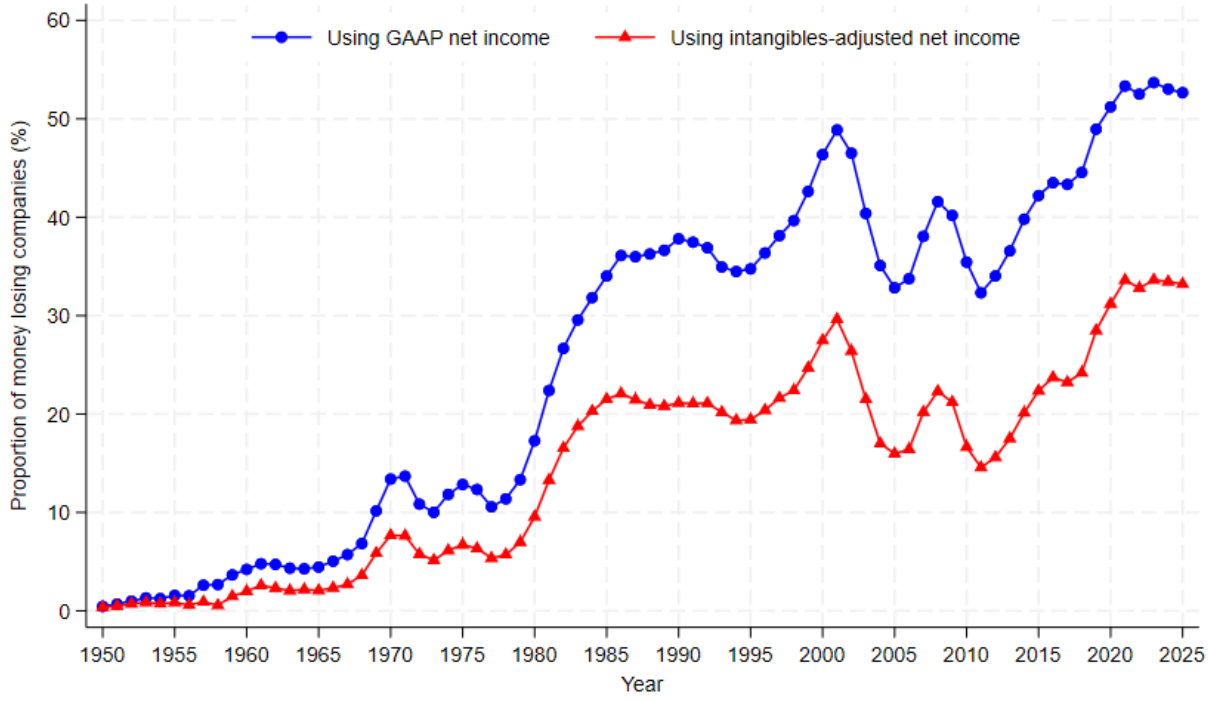
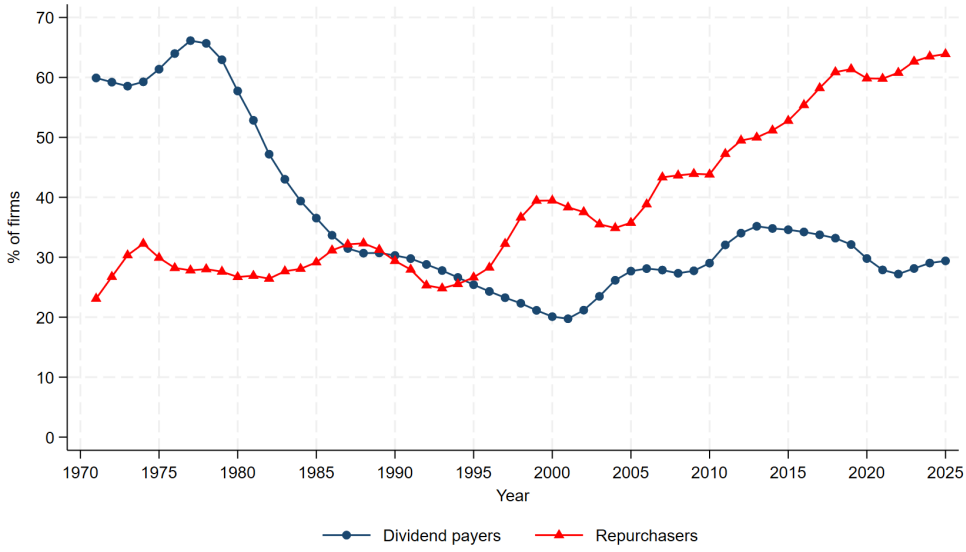


Figure 9: Payouts

Panel (a) plots the share of firms that pay dividends and the share of firms that repurchase shares; panel (b) plots aggregate dividends and aggregate repurchases relative to aggregate total assets, stacked, so that the top of the shaded area is total payouts. Three-year centered moving averages, 1971–2025.

(a) Dividend payers and repurchasers



(b) Payouts relative to total assets

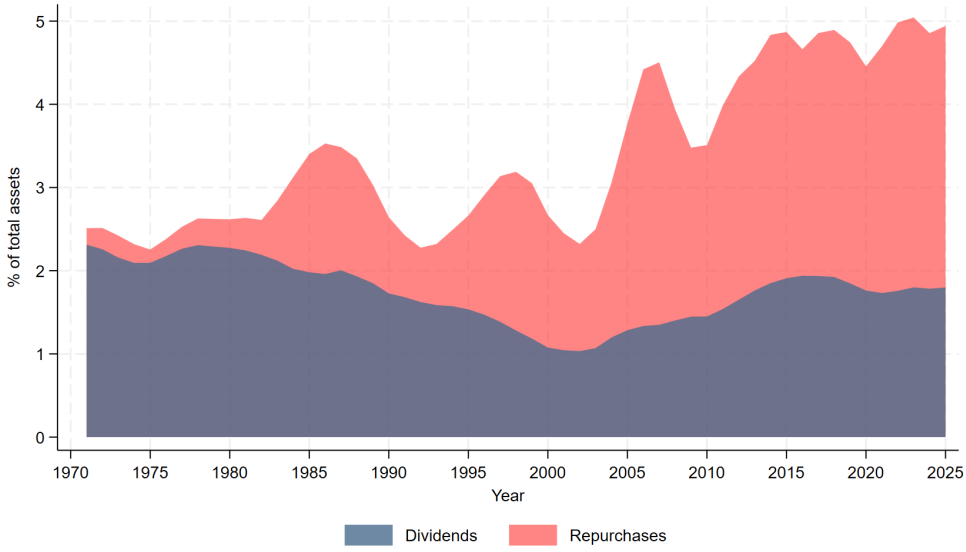


Figure 10: Write-downs and impairments

This figure plots the number of firms with a write-down or impairment (left axis) and the average write-down relative to assets among them (right axis), from 1995.

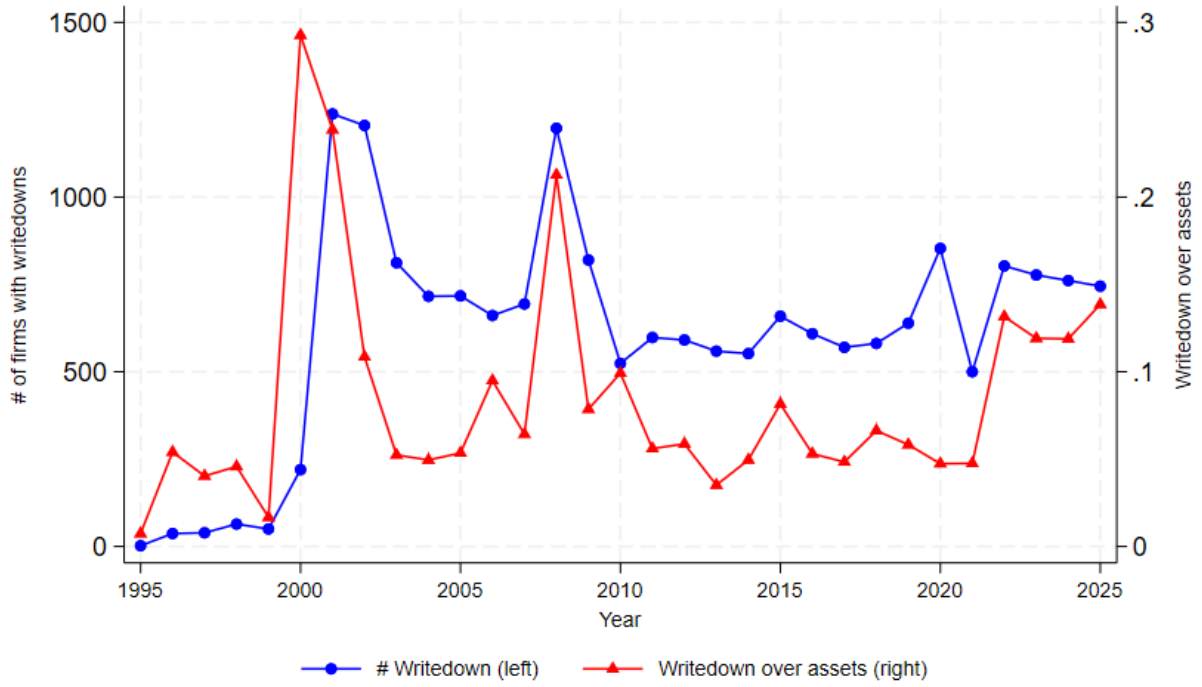


Figure 11: Intangible investment

This figure plots intangible investment (R&D plus 30% of SG&A) relative to capital expenditures (left axis) and relative to tangible book assets (right axis), three-year moving averages.

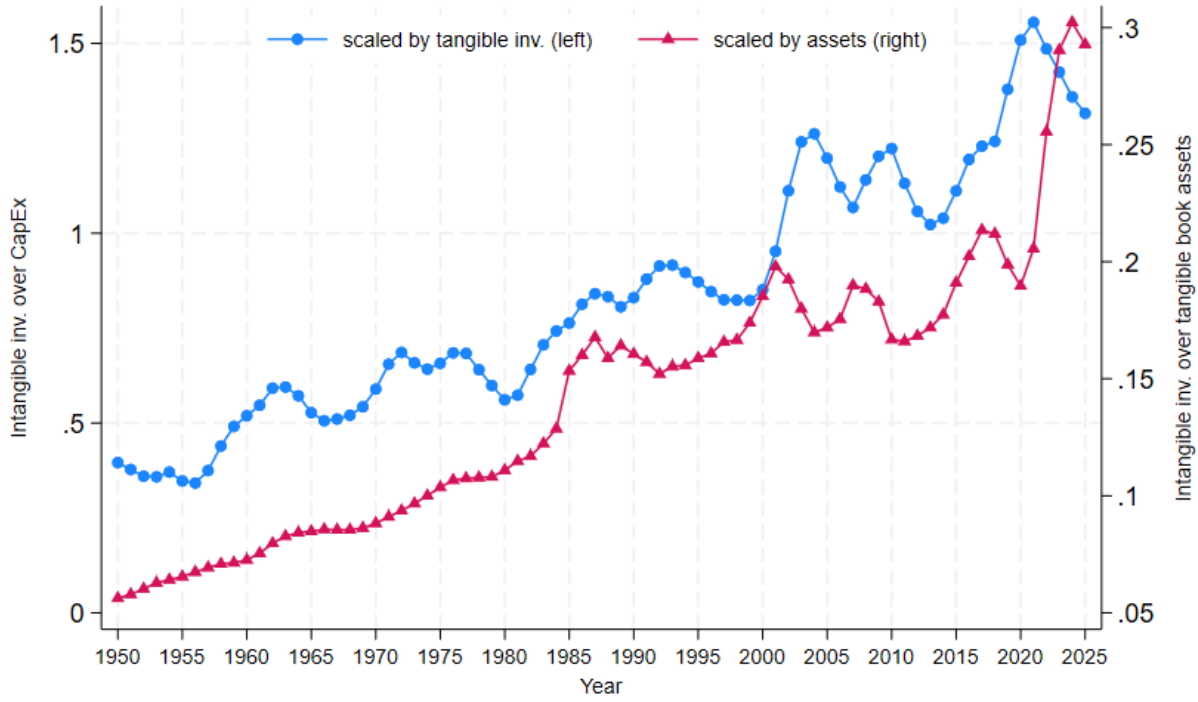


Figure 12: The anatomy of red equity

This figure plots the share of red equity firms among all firms and the share that would remain red equity after adjusting book equity by (a) adding back cumulative payouts (dividends and repurchases), (b) adding back cumulative write-downs and impairments, (c) adding intangible capital, and (d) all three adjustments together.

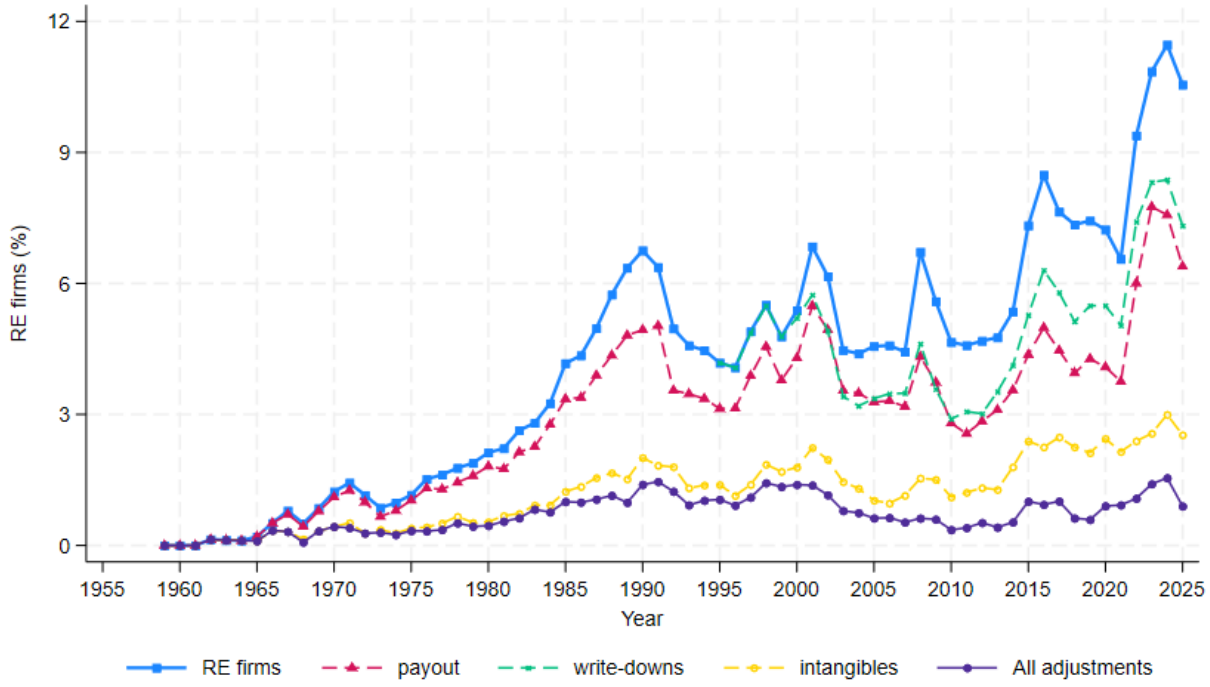


Figure 13: Two kinds of red equity

Panel (a) plots the share of all firms that are red equity firms, by source of the deficit; panel (b) plots the split of the market value of red equity firms by source. Payout-driven: lifetime earnings (retained earnings in the first Compustat record plus net income since) are non-negative; write-down-driven: negative, but non-negative after adding back cumulative write-downs; operating-loss-driven: otherwise.

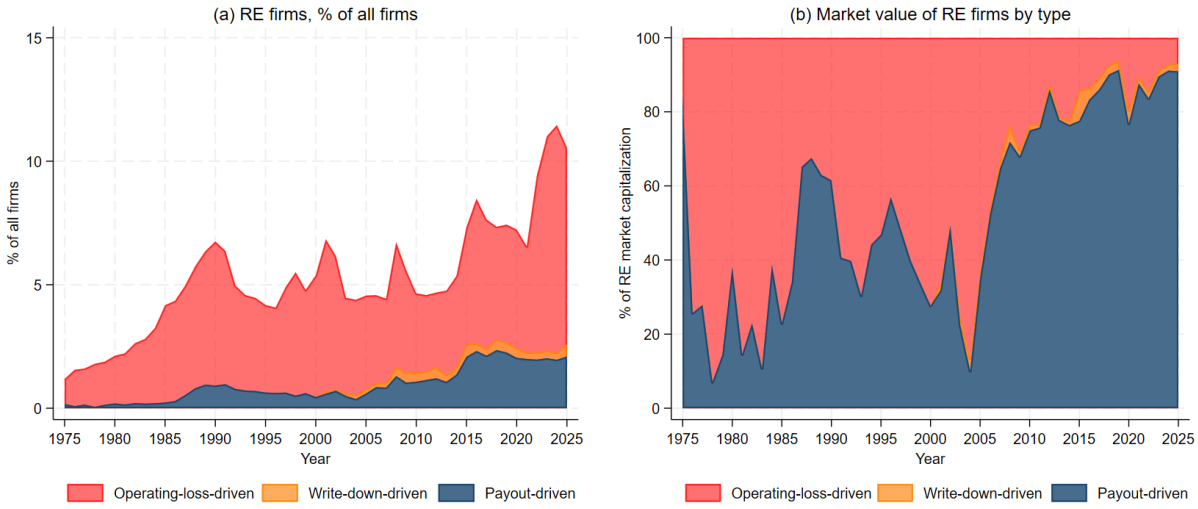


Figure 14: Economic weight of payout-driven red equity firms

This figure plots the share of payout-driven firms in the market capitalization, total assets, and employment of all red equity firms (three-year centered moving averages). Their share in the number of red equity firms is reported in Table 5.



Figure 15: Including red equity firms in intangible-adjusted HML: cumulative abnormal returns

This figure plots, from January 1970 to August 2026, the cumulative sum of the monthly return difference between HML_INT inc RE and HML_INT ex RE (raw difference) and of its CAPM and FF abnormal returns, in percentage points. The abnormal return in a month is the difference minus its factor exposures, that is, the alpha plus the residual of a regression of the monthly difference on the market factor (CAPM) or on the market, size, profitability, and investment factors of Fama and French (2015) (FF), with loadings estimated over the full sample. The average slope of each line equals the corresponding average return or alpha in Table 12, Panel A, Column 5.

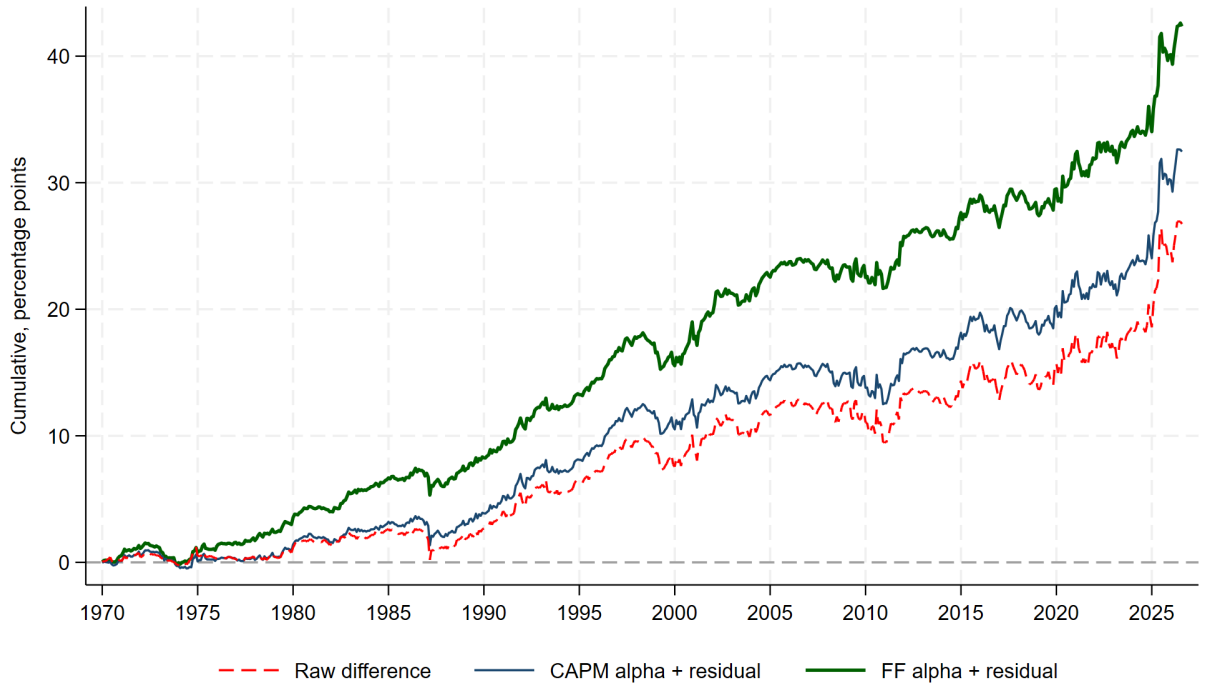


Figure 16: Including red equity firms in intangible-adjusted HML, by start of the sample

This figure plots the difference between HML_INT inc RE and HML_INT ex RE (Table 12) in average return, CAPM alpha, and FF alpha, in percent per year, for samples that start in July of the year shown and end with the data. The CAPM alpha is the intercept of a regression of the monthly difference on the market excess return; the FF alpha is the intercept of a regression on the market, size, profitability, and investment factors of Fama and French (2015), as in Table 12. The shaded area is the 95% confidence interval of the FF alpha, based on Newey–West standard errors with one lag.

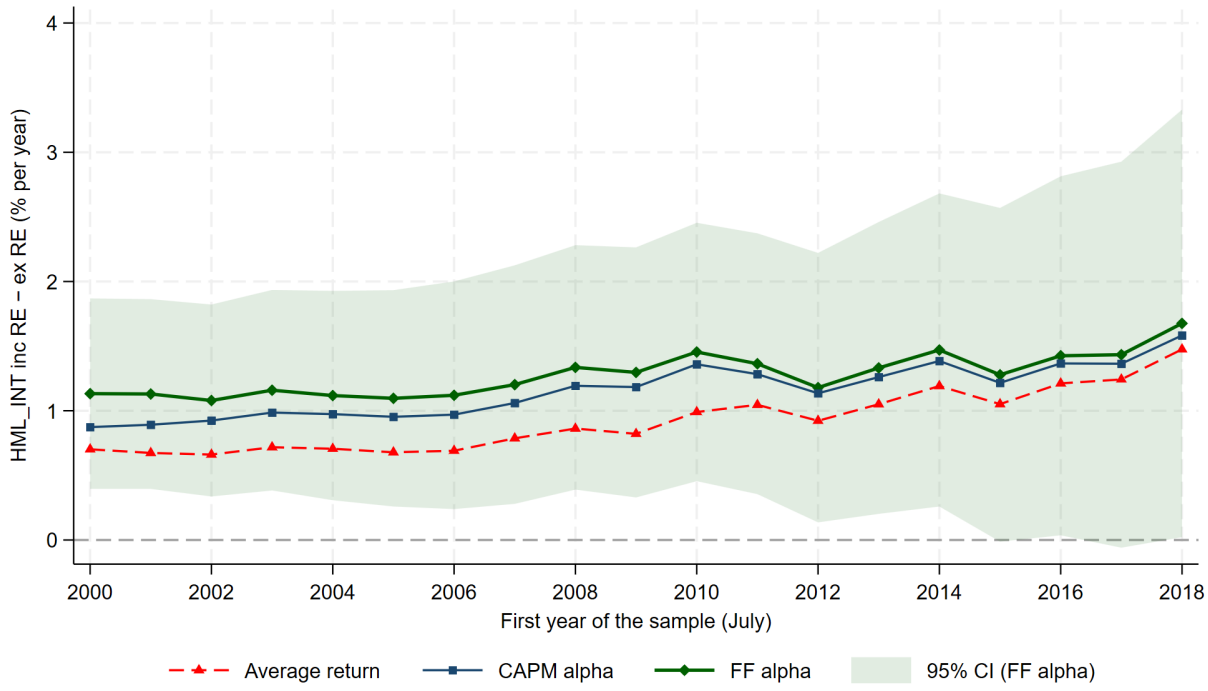


Figure 17: Red equity across countries

This figure plots the share of listed firms with negative common equity in the United States (main sample) and in other countries (Compustat Global, industrial format, consolidated statements, exchange-listed firms, excluding utilities and financial firms). Europe pools all listed firms incorporated in the 27 EU member states, the United Kingdom, Switzerland, Norway, and Iceland. The individual countries shown have at least 100 such firms in every year 1995–2025; other countries are reported in Table 14 and Table A5. India is excluded (see Section 8). Common equity is reported under each firm's accounting standards.

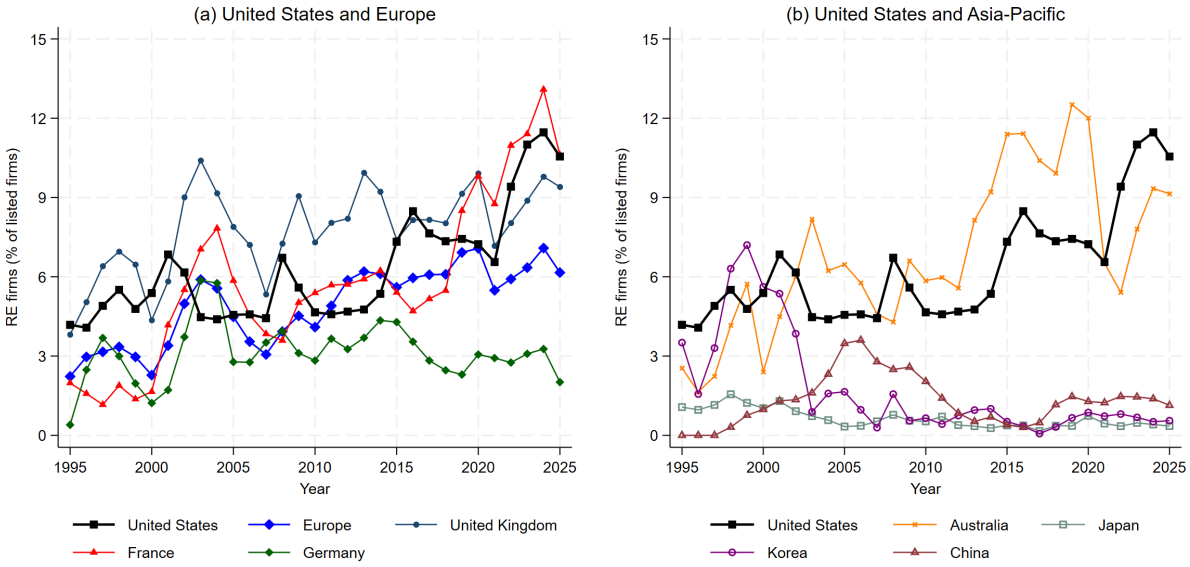


Table 1: Descriptive statistics

This table reports the descriptive statistics for the main variables in this paper. *RedEquity* is an indicator variable for having negative common equity. *LossMaking* is an indicator variable for whether a firm has negative net income. *Intan_over_Assets* is the ratio of intangible investments to total assets. *Repurchaser* and *DividendPayer* are indicator variables for whether a firm repurchased or paid dividend, respectively. *Repurchase* and *Dividend* are log of respective variables. *Repurchase_over_Assets*, *Dividend_over_Assets*, and *Writedown_over_Assets* are the ratio of repurchases, dividends, and writedown and impairments, respectively, to total assets. *Size* is the log market capitalization. *Leverage* is total debt over assets. *Age* is age since IPO. *Profitability* is the ratio of earnings before interest, taxes and depreciation to total assets. *Tax* is the ratio of taxes paid to total assets. *Annual_Return* is the compounded monthly returns over the year. All variables are winsorized at the 1% level in each tail. Table A1 presents detailed variable definitions.

Variables	N	Mean	SD	Min	p50	Max
RedEquity	223,283	0.05	0.21	0.00	0.00	1.00
LossMaking	222,868	0.33	0.47	0.00	0.00	1.00
Intan_over_Assets	200,728	0.13	0.13	0.00	0.09	0.75
Repurchase_over_Assets	207,603	0.01	0.03	0.00	0.00	0.22
Dividend_over_Assets	223,283	0.01	0.02	0.00	0.00	0.11
Payout_over_Assets	223,283	0.02	0.04	0.00	0.00	0.27
Writedown_over_Assets	114,452	0.01	0.04	0.00	0.00	0.33
Size	223,283	4.76	2.33	0.23	4.56	10.75
Leverage	221,808	0.24	0.22	0.00	0.21	0.96
TobinQ	221,768	1.72	1.76	0.29	1.12	11.19
Age	223,283	12.02	12.31	0.00	8.00	56.00
Profitability	222,272	0.05	0.26	-1.30	0.11	0.41
Tax	222,876	0.03	0.04	-0.08	0.02	0.16
Annual_Return	223,283	0.12	0.63	-0.87	0.03	3.00

Table 2: Red equity firms

This table reports the number, proportion and total market capitalization (in $ bn) of firms with negative equity over the years.

Year	Number	Proportion	Market Cap
1961	0	0	0
1962	3	0.38	0.04
1963	3	0.34	0.05
1964	3	0.31	0.06
1965	6	0.55	0.05
1966	10	0.77	0.12
1967	16	1.14	1.05
1968	10	0.65	2.12
1969	16	0.95	1.43
1970	23	1.26	1.54
1971	29	1.51	1.90
1972	38	1.38	1.86
1973	31	0.96	1.32
1974	33	0.97	1.09
1975	40	1.18	1.39
1976	53	1.56	0.52
1977	54	1.61	0.84
1978	60	1.80	1.15
1979	65	1.89	1.49
1980	75	2.12	1.64
1981	85	2.22	1.67
1982	101	2.63	3.73
1983	119	2.80	4.25
1984	141	3.25	5.45
1985	179	4.17	5.20
1986	194	4.35	8.41
1987	231	4.97	14.71
1988	256	5.74	27.65
1989	272	6.36	25.73
1990	286	6.75	21.09
1991	275	6.37	25.43
1992	221	4.97	34.99
1993	223	4.58	62.60
1994	230	4.47	45.62
1995	223	4.18	55.64
1996	236	4.07	60.58
1997	285	4.90	97.93
1998	300	5.51	105.68
1999	249	4.78	136.10
2000	271	5.39	100.57
2001	303	6.84	123.76
2002	251	6.16	104.92
2003	169	4.47	117.92
2004	165	4.39	114.38
2005	168	4.56	105.11
2006	166	4.58	180.66
2007	159	4.43	125.54
2008	227	6.72	155.76
2009	178	5.59	154.56
2010	143	4.65	237.51
2011	136	4.58	204.94
2012	135	4.68	340.41
2013	138	4.76	420.83
2014	161	5.35	476.96
2015	218	7.33	627.85
2016	245	8.48	759.76
2017	219	7.64	908.99
2018	212	7.35	1,175.62
2019	214	7.44	1,570.27
2020	216	7.23	1,240.88
2021	220	6.56	2,499.99
2022	315	9.41	1,714.57
2023	353	11.00	1,994.28
2024	356	11.47	2,141.61
2025	317	10.55	2,620.26

Table 3: Descriptive statistics by RedEquity

This table reports the average of main variables, separately for positive equity and negative equity firms. All the differences are statistically significant. All variables are defined in the notes to Table 1. Detailed definitions are also provided in Table A1.

	RedEquity=0	RedEquity=1	Diff (1 − 0)
LossMaking	0.310	0.759	0.449
Profitability	0.059	-0.214	-0.273
Intan_over_Assets	0.124	0.216	0.092
Repurchase_over_Assets	0.012	0.014	0.002
Dividend_over_Assets	0.010	0.005	-0.005
Payout_over_Assets	0.022	0.021	-0.001
Writedown_over_Assets	0.008	0.024	0.016
Size	4.791	4.111	-0.681
Leverage	0.222	0.608	0.386
TobinQ	1.668	2.711	1.044
Age	12.054	11.255	-0.798

Table 4: Determinants of red equity

This table presents the relationship between firm characteristics and the red equity indicator. The dependent variable is a binary indicator equal to 1 if a firm has negative equity in a given year, and 0 otherwise. For presentation purposes, the regression estimates are multiplied by 100 to improve numerical readability in the table. All non-binary independent variables are standardized to have the mean 0 and standard deviation of 1. Standard errors, clustered at the firm level, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. All variables are defined in Table A1.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	RedEquity					
LossMaking	6.48***	6.67***	4.83***	1.59***	0.90***	0.51**
	(0.23)	(0.23)	(0.32)	(0.24)	(0.25)	(0.25)
Profitability	-1.82***	-2.06***	-2.60***	-1.27***	-1.44***	-1.58***
	(0.15)	(0.14)	(0.30)	(0.24)	(0.24)	(0.24)
Intan_over_Assets	0.94***	0.83***	0.95***	2.99***	2.99***	3.18***
	(0.11)	(0.11)	(0.23)	(0.21)	(0.21)	(0.22)
Payout_over_Assets		0.86***	1.35***	1.34***	1.32***	1.23***
		(0.11)	(0.19)	(0.13)	(0.13)	(0.12)
Writedown_over_Assets			1.01***	0.64***	0.67***	0.61***
			(0.14)	(0.11)	(0.11)	(0.10)
Size				-2.72***	-2.79***	-2.96***
				(0.16)	(0.16)	(0.16)
Leverage				11.43***	11.44***	11.43***
				(0.26)	(0.26)	(0.25)
Age				0.49***	0.46***	0.65***
				(0.16)	(0.16)	(0.17)
TobinQ				3.43***	3.40***	3.45***
				(0.17)	(0.16)	(0.16)
Observations	200,369	200,369	101,518	101,097	101,097	101,097
R-squared	0.05	0.05	0.04	0.26	0.27	0.28
Year FE	No	No	No	No	Yes	Yes
Industry FE	No	No	No	No	No	Yes

Table 5: Types of red equity firms

RE firms are classified by the source of their negative book equity, using lifetime earnings (retained earnings in the first year in which the firm reports them, plus the dividends paid in that year and in earlier records, plus all net income since). Payout-driven: lifetime earnings are positive, so equity is negative because payouts exceed lifetime earnings. Write-down-driven: lifetime earnings are negative but become positive after adding back cumulative write-downs (goodwill impairments and other pre-tax write-downs, available from 1995). Operating-loss-driven: lifetime earnings are negative even after adding back write-downs. Shares of RE firm-years and of RE market capitalization are pooled over the panel years (market capitalization in nominal dollars, so recent years weigh more). Distressed: at least two of Altman Z below 1.81, naive default probability (Bharath and Shumway, 2008) above 10%, and EBITDA below interest expense; int.-adj.: Altman Z and interest coverage adjusted for expensed intangible investment. Delisted: performance-related delisting from CRSP within 24 months. Large: market capitalization above the yearly median. Positive with int. capitalized: share of RE firm-years whose common equity plus off-balance-sheet intangible capital is non-negative (firm-years with available intangible capital). Panel A covers 1975–2025 and Panel B 2010–2025.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
	Firm-years	% of RE firm-years	% of RE market cap	Distressed reported	(2 of 3) int.-adj.	Default prob. > 10%	Delisted (24 m)	Still RE after 1 yr	3 yrs	5 yrs	Large (%)	Years since IPO	Profitable (%)	Positive with int. capitalized (%)
Panel A: 1975–2025														
Non-RE firms	186,186			15.0	9.4	10.6	4.1				51.2	13.0	65.7	
RE: payout-driven	1,689	16.7	82.6	22.7	18.8	23.8	6.6	79.2	59.8	47.0	68.3	23.2	69.3	67.7
RE: write-down-driven	237	2.3	2.2	61.4	58.3	57.9	16.4	83.9	62.3	48.1	34.2	23.5	33.8	57.4
RE: operating-loss-driven	8,182	80.9	15.2	79.0	65.6	49.7	26.7	70.0	45.3	33.5	20.1	8.8	13.8	71.9
Panel B: 2010–2025														
Non-RE firms	44,733			19.3	12.4	9.3	2.3				51.4	18.9	57.1	
RE: payout-driven	880	24.5	86.7	16.0	13.2	18.1	3.7	84.4	69.3	60.2	71.0	25.1	78.0	65.5
RE: write-down-driven	175	4.9	2.3	61.4	59.0	56.7	13.2	83.8	64.2	53.8	33.7	27.0	37.1	58.3
RE: operating-loss-driven	2,543	70.7	11.0	79.7	67.1	46.0	17.8	71.0	50.9	41.3	19.2	10.2	13.9	75.1

Table 6: Types of red equity firms and intangible-adjusted equity

This table classifies RE firm-years by the source of their deficit (Table 5; loss-driven combines write-down- and operating-loss-driven firms) and by the sign of their equity once internally generated intangibles are capitalized (common equity plus off-balance-sheet intangible capital). Shares of RE firm-years and of RE market capitalization are relative to all RE firm-years and are pooled over the panel years; the few RE firm-years without intangible capital are not shown. Distressed: at least two of Altman Z below 1.81, naive default probability (Bharath and Shumway, 2008) above 10%, and EBITDA below interest expense; int.-adj.: Altman Z and interest coverage adjusted for expensed intangible investment. Delisted: performance-related delisting from CRSP within 24 months. Profitable: positive net income. Large: market capitalization above the yearly median. Panel A covers 1975–2025 and Panel B 2010–2025.

	(1) Firm-years	(2) % of RE firm-years	(3) % of RE market cap	(4) Distressed reported	(5) (2 of 3) int.-adj.	(6) Delisted (24 m)	(7) Profitable (%)	(8) Large (%)
Panel A: 1975–2025								
Payout-driven: positive with intangibles	1,144	11.3	56.9	24.0	19.0	6.5	65.4	67.9
Payout-driven: negative with intangibles	545	5.4	25.7	19.8	18.2	6.9	77.5	69.2
Loss-driven: positive with intangibles	6,008	59.4	11.2	79.1	62.4	26.0	12.8	17.9
Loss-driven: negative with intangibles	2,399	23.7	6.2	77.0	72.6	27.7	18.4	26.8
Panel B: 2010–2025								
Payout-driven: positive with intangibles	576	16.0	59.2	19.0	14.7	4.0	73.8	70.5
Payout-driven: negative with intangibles	304	8.4	27.4	10.2	10.2	3.0	85.9	72.0
Loss-driven: positive with intangibles	2,007	55.8	8.6	80.0	65.4	16.2	12.4	18.5
Loss-driven: negative with intangibles	704	19.6	4.7	74.0	69.6	20.8	24.0	24.7

Table 7: Financial distress, delisting and persistence of red equity firms

This table reports, for each group of firm-years, the percentage classified as distressed and the percentage still RE one, three and five years later. Distressed: at least two of (i) Altman (1968) Z-score below 1.81, (ii) naive default probability (Bharath and Shumway, 2008) above 10%, (iii) EBITDA below interest expense. Int.-adj.: the same composite with the Altman Z-score and interest coverage adjusted for expensed intangible investment (intangible investment added back to EBIT and EBITDA, intangible capital to retained earnings and assets). Delisting: performance-related delisting from CRSP within 24 months after the fiscal year end. RE subgroups are based on book equity adjusted for intangible capital (ptx measure), repurchases and write-downs. Size groups use the yearly median market capitalization. Panel A covers 1975–2025 and Panel B 2010–2025.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Firm-years	Distressed reported	(2 of 3) int.-adj.	Altman Z < 1.81	Default prob. $> 10\%$	EBITDA $<$ interest	Delisted (24 m)	Still RE after 1 yr	3 yrs	5 yrs
Panel A: 1975–2025										
Non-RE firms	186,186	15.0	9.4	22.4	10.6	24.8	4.1			
RE firms (all)	10,108	68.8	57.3	81.6	45.5	62.6	23.2	72.2	49.1	37.2
RE firms above median size	2,876	34.4	26.0	62.8	19.4	30.9	5.2	76.7	57.3	45.5
RE firms below median size	7,232	82.9	70.2	89.3	55.8	75.5	30.4	69.8	43.6	31.1
RE: positive after intangibles	7,152	69.7	55.1	80.4	43.1	65.6	22.9	67.1	43.5	31.5
RE: positive after payout/write-downs	936	50.0	46.9	75.7	44.0	34.5	17.5	88.9	71.0	58.7
RE: negative after all adjustments	2,008	74.1	69.9	88.8	55.1	65.2	26.7	82.4	56.9	44.3
Panel B: 2010–2025										
Non-RE firms	44,733	19.3	12.4	32.1	9.3	31.1	2.3			
RE firms (all)	3,598	62.4	52.9	78.6	39.6	56.4	13.9	75.3	57.7	48.6
RE firms above median size	1,171	22.5	15.6	51.1	11.0	21.1	1.0	82.0	65.2	54.5
RE firms below median size	2,427	83.1	72.2	93.4	53.4	73.9	20.5	71.3	51.8	43.7
RE: positive after intangibles	2,583	65.4	53.3	79.6	37.9	61.5	13.4	69.9	50.2	40.6
RE: positive after payout/write-downs	519	41.6	39.4	66.7	39.1	26.6	10.9	91.3	79.3	70.7
RE: negative after all adjustments	489	69.1	64.9	86.5	49.6	60.8	19.9	86.6	70.0	61.8

Table 8: Leverage regressions with RedEquity interactions

This table presents leverage regressions estimated on the pooled sample of RE and non-RE firms. The dependent variable is leverage (total debt over total assets), multiplied by 100 and winsorized at the 1st and 99th percentiles separately for RE and non-RE firms. All right-hand-side variables are lagged by one year and standardized to have a mean of 0 and a standard deviation of 1 each year. *RE* is an indicator equal to 1 if the firm has negative common equity in the current year. The coefficients on the determinants measure their effect for non-RE firms, and the coefficients on the interactions with *RE* measure the difference for RE firms. Standard errors, clustered at the firm level, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. All variables are defined in Table A1.

	(1)	(2)	(3)
Dependent variable:		Leverage	
Log_Assets	3.33***	3.32***	3.32***
	(0.14)	(0.14)	(0.14)
Profitability	-0.24**	-0.13	-0.31***
	(0.12)	(0.12)	(0.11)
Tangibility	2.12***	2.15***	3.01***
	(0.14)	(0.13)	(0.16)
Ind_Leverage	4.46***	4.48***	4.41***
	(0.14)	(0.13)	(0.17)
TobinQ	-0.83***	-0.86***	-0.70***
	(0.09)	(0.09)	(0.09)
Intan_over_Assets	-1.89***	-1.85***	-2.22***
	(0.11)	(0.11)	(0.12)
Tax	-3.92***	-3.97***	-4.09***
	(0.09)	(0.09)	(0.09)
Age	-1.46***	-1.47***	-1.55***
	(0.11)	(0.11)	(0.11)
Annual_Return	0.09*	0.09*	0.12***
	(0.05)	(0.05)	(0.04)
RE	47.01***	47.04***	46.97***
	(1.20)	(1.20)	(1.19)
RE × Log_Assets	0.95	1.11	0.94
	(1.16)	(1.16)	(1.15)
RE × Profitability	1.20	1.16	1.16
	(1.13)	(1.14)	(1.13)
RE × Tangibility	2.02**	2.14**	2.07**
	(0.93)	(0.93)	(0.91)
RE × Ind_Leverage	0.29	0.29	0.20
	(1.01)	(1.01)	(1.00)
RE × TobinQ	8.54***	8.46***	8.34***
	(1.61)	(1.62)	(1.61)
RE × Intan_over_Assets	-0.78	-0.51	-0.45
	(1.03)	(1.03)	(1.02)
RE × Tax	7.20***	7.28***	7.34***
	(1.28)	(1.28)	(1.29)
RE × Age	-0.73	-0.70	-0.43
	(0.93)	(0.93)	(0.93)
RE × Annual_Return	-2.88***	-2.81***	-2.89***
	(0.62)	(0.62)	(0.62)
Observations	180,430	180,430	180,430
R-squared	0.34	0.36	0.37
Year FE	No	Yes	Yes
Industry FE	No	No	Yes

Table 9: Leverage regressions by type of red equity firm

This table presents leverage regressions on the pooled sample of RE and non-RE firms, with the determinants interacted with indicators for payout-driven and loss-driven (write-down- or operating-loss-driven) RE firms (see Table 5). The dependent variable is leverage (total debt over total assets), multiplied by 100 and winsorized at the 1st and 99th percentiles separately for RE and non-RE firms. A single regression of leverage on the determinants, their interactions with indicators for payout-driven and loss-driven RE firm-years, and the two indicators is estimated. Column 1 reports the coefficients on the determinants, which measure their effect for non-RE firms; Columns 2 and 3 report the coefficients on the interactions, which measure the difference in the effect for payout-driven and loss-driven RE firms, so that the effect for these firms is the sum of the two coefficients. The coefficients on the two indicators are reported in the last rows. All right-hand-side variables are lagged by one year and standardized to have a mean of 0 and a standard deviation of 1 each year. All regressions include year and industry (Fama-French 49) fixed effects. Standard errors, clustered at the firm level, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	Leverage		
	(1)	(2)	(3)
	Non-RE	× Payout	× Loss
Log_Assets	3.32***	-10.62***	5.66***
	(0.14)	(2.42)	(1.28)
Profitability	-0.31***	14.93***	-1.09
	(0.11)	(3.25)	(1.00)
Tangibility	3.02***	1.69	1.99**
	(0.16)	(1.83)	(0.97)
Ind_Leverage	4.43***	3.76*	-1.75*
	(0.17)	(2.11)	(0.94)
TobinQ	-0.70***	16.60***	6.56***
	(0.09)	(4.83)	(1.35)
Intan_over_Assets	-2.22***	1.75	-0.87
	(0.12)	(3.91)	(1.02)
Tax	-4.09***	2.73*	6.74***
	(0.09)	(1.47)	(1.10)
Age	-1.56***	0.32	0.90
	(0.11)	(1.65)	(1.09)
Annual_Return	0.12***	-2.31**	-2.39***
	(0.04)	(0.98)	(0.63)
RE: payout-driven (indicator)	40.96***		
	(2.75)		
RE: loss-driven (indicator)	49.69***		
	(1.38)		
Observations		180,430	
R-squared		0.38	

Table 10: Firms with negative tangible equity

This table compares firms with non-negative tangible equity, firms with non-negative book equity but negative tangible equity, and RE firms. Tangible equity is common equity minus intangible assets reported on the balance sheet (INTAN, including goodwill), which are almost entirely acquired; it equals book equity if acquired intangibles were expensed like internally generated ones. Shares of firm-years and of market capitalization are pooled over the panel years (market capitalization in nominal dollars). Acquired intangibles: INTAN over total assets; goodwill share: goodwill over INTAN. Distressed: at least two of Altman Z below 1.81, naive default probability above 10%, and EBITDA below interest expense; int.-adj.: with the accounting-based measures adjusted for expensed intangible investment. Delisted: performance-related delisting from CRSP within 24 months. Large: market capitalization above the yearly median. All figures except firm-years are in percent. Panel A covers 1975–2025 and Panel B 2010–2025.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	Firm-years	% of firm-years	% of market cap	Acquired intangibles	Goodwill share	Book leverage	Profit-able	Large	Distressed reported	Distressed int.-adj.	Delisted (24 m)
Panel A: 1975–2025											
Tangible equity ≥ 0	169,154	86.2	72.8	7.0	63.4	20.1	66.2	49.8	13.9	8.5	3.9
Book equity ≥ 0, tangible equity < 0	17,032	8.7	23.8	46.5	65.8	41.1	60.4	64.9	25.1	18.8	6.2
Book equity < 0 (RE firms)	10,108	5.1	3.4	12.7	53.9	74.9	23.6	28.5	68.8	57.3	23.2
Panel B: 2010–2025											
Tangible equity ≥ 0	35,703	73.9	66.9	12.2	54.8	17.4	55.2	46.6	19.1	11.8	2.3
Book equity ≥ 0, tangible equity < 0	9,030	18.7	28.7	50.2	61.3	39.6	64.7	70.5	19.8	14.7	2.4
Book equity < 0 (RE firms)	3,598	7.4	4.4	17.5	50.7	83.1	30.7	32.5	62.4	52.9	13.9

Table 11: Value relevance of reported and intangible-adjusted book value

This table reports regressions of the share price on book value and earnings per share, estimated separately for firms with positive book equity, payout-driven RE firms and loss-driven (write-down- or operating-loss-driven) RE firms (see Table 5). In Panel A, book value is common equity and earnings are income before extraordinary items. In Panel B, book value is intangible-adjusted: common equity plus intangible capital; earnings are adjusted accordingly: income before extraordinary items plus intangible investment minus the amortization of intangible capital implied by the capital stock, net of taxes at the firm's effective tax rate. Intangible capital is added at its pre-tax amount. Firms are classified by their reported book equity in both panels. Price is market value divided by shares outstanding. Both panels use the firm-years with intangible capital in the current and the previous year, 1975–2025. All regressions include year and industry (Fama-French 49) fixed effects. Variables are winsorized at the 1% level in each tail. Standard errors, clustered at the firm level, are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The last two rows of each panel report the p-values of tests that the book-value coefficient of payout-driven RE firms equals that of positive-equity firms, (2) = (1), and that of loss-driven RE firms, (2) = (3), from a pooled regression in which book value, earnings, and the year and industry fixed effects are interacted with indicators for the two types of RE firms, so that its coefficients equal those of the separate regressions.

	(1)	(2)	(3)
	Positive	RE firms	
	book equity	Payout-driven	Loss-driven
Panel A: reported book value and earnings			
Book value	1.14***	-3.59***	-1.83***
	(0.03)	(0.76)	(0.45)
Earnings	4.90***	7.59***	0.98**
	(0.14)	(0.72)	(0.46)
Observations	166,159	1,575	7,611
R-squared	0.60	0.62	0.14
p-value, book value: (2) = (1)		< 0.001	
p-value, book value: (2) = (3)		0.041	
Panel B: intangible-adjusted book value and earnings			
Adjusted book value	0.51***	0.62***	0.24***
	(0.02)	(0.10)	(0.04)
Adjusted earnings	5.46***	7.09***	1.17***
	(0.11)	(0.57)	(0.38)
Observations	166,159	1,575	7,611
R-squared	0.60	0.67	0.13
p-value, book value: (2) = (1)		0.284	
p-value, book value: (2) = (3)		< 0.001	

Table 12: Intangible-adjusted HML and red equity firms

This table reports annualized average returns and alphas (in %) of four value-weighted HML strategies that are constructed identically except for the book equity used and the firms included. HML uses reported common equity and excludes firms with negative values. The three HML_INT strategies use intangible-adjusted common equity (common equity plus off-balance-sheet intangible capital): HML_INT excludes firms with negative intangible-adjusted equity, HML_INT ex RE excludes firms with negative reported common equity, and HML_INT inc RE includes all firms. Column (5) is the difference between columns (4) and (3). CAPM alpha is the intercept of a regression on the market excess return; FF alpha is the intercept of a regression on the market, size, profitability, and investment factors of Fama and French (2015). Newey–West t-statistics with one lag are in parentheses. Panel A covers January 1970 to August 2026, Panel B July 2009 to August 2026.

	(1) HML	(2) HML_INT	(3) HML_INT ex RE	(4) HML_INT inc RE	(5) Difference (4) − (3)
Panel A: full sample					
Average return	6.64	9.13	8.86	9.33	0.47
	(3.83)	(5.37)	(5.19)	(5.61)	(2.55)
CAPM alpha	7.57	8.91	8.58	9.15	0.57
	(4.26)	(5.12)	(4.93)	(5.37)	(3.24)
FF alpha	2.03	4.83	4.50	5.25	0.75
	(1.34)	(3.57)	(3.35)	(3.95)	(3.89)
Panel B: July 2009–2026					
Average return	5.10	8.36	8.11	8.93	0.82
	(1.54)	(2.50)	(2.40)	(2.76)	(1.62)
CAPM alpha	4.61	5.94	5.54	6.72	1.18
	(1.26)	(1.71)	(1.59)	(1.99)	(2.45)
FF alpha	4.20	6.60	6.11	7.41	1.30
	(1.64)	(2.67)	(2.45)	(3.04)	(2.63)

Table 13: Intangible-adjusted HML and the types of red equity firms

This table reports average annualized returns, CAPM alphas, and FF alphas (in %) of value-weighted intangible-adjusted HML strategies (ptx measure) that differ only in which RE firms are included: none (ex RE), all (inc RE), only payout-driven RE firms, or only write-down- and operating-loss-driven RE firms (see Table 5). Columns 1 and 2 repeat Columns 3 and 4 of Table 12. FF alpha is the intercept of a regression on the market, size, profitability, and investment factors of Fama and French (2015). Differences are relative to the strategy without RE firms. Newey–West t-statistics with one lag are in parentheses. Panel A covers 1970m1 to 2026m8; Panel B starts in July 2009.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	HML_INT including RE firms:				Difference to ex RE		
	none	all	payout-driven	loss-driven	all	payout-driven	loss-driven
Panel A: full sample							
Average return	8.86	9.33	8.95	9.16	0.47	0.09	0.31
	(5.19)	(5.61)	(5.25)	(5.45)	(2.55)	(1.04)	(2.00)
CAPM alpha	8.58	9.15	8.68	8.99	0.57	0.10	0.41
	(4.93)	(5.37)	(4.99)	(5.23)	(3.24)	(1.19)	(2.71)
FF alpha	4.50	5.25	4.74	4.97	0.75	0.24	0.47
	(3.35)	(3.95)	(3.51)	(3.71)	(3.89)	(2.70)	(2.75)
Panel B: July 2009–2026							
Average return	8.11	8.93	8.37	8.65	0.82	0.26	0.54
	(2.40)	(2.76)	(2.48)	(2.62)	(1.62)	(1.09)	(1.33)
CAPM alpha	5.54	6.72	5.89	6.49	1.18	0.36	0.95
	(1.59)	(1.99)	(1.69)	(1.89)	(2.45)	(1.50)	(2.41)
FF alpha	6.11	7.41	6.52	7.15	1.30	0.42	1.04
	(2.45)	(3.04)	(2.59)	(2.91)	(2.63)	(1.72)	(2.50)

Table 14: Red equity outside the United States

This table reports the share of listed firms with negative common equity (RE firms), the same share excluding the smallest 20% of firms by total assets in each country and year, the share that would have negative equity if all intangible assets on the balance sheet were expensed (tangible equity, common equity minus intangible assets including goodwill, below zero), and the share of RE firm-years that are payout-driven (lifetime earnings non-negative). The last column is left blank for countries with fewer than 30 RE firm-years over 2015–2025, for which the share is not meaningful. Data for countries other than the United States come from Compustat Global (industrial format, consolidated statements, exchange-listed firms, excluding utilities, banks, and insurance companies, as in the US sample); common equity is reported under each firm's accounting standards. Europe pools all listed firms incorporated in the 27 EU member states, the United Kingdom, Switzerland, Norway, and Iceland. The table reports every European country with at least 1,000 listed firm-years over 2015–2025 and the five largest other markets; Table A5 reports the remaining non-European countries with at least 1,000 listed firm-years. India is excluded because of a break in its series in 2011, when most Indian firms with negative equity had positive equity a year earlier; from fiscal 2011–12, India's Revised Schedule VI required accumulated losses previously shown on the assets side of the balance sheet to be deducted from shareholders' funds. Firms incorporated in Bermuda, the Cayman Islands, and the British Virgin Islands are also excluded. Data for the United States come from the main sample. Averages are pooled over 2015–2025.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Firm-years	RE firms (%)		RE firms excl. smallest 20% (%)		Negative tangible equity (%)		Payout-driven
	2015–2025	2015–2025	2025	2015–2025	2025	2015–2025	2025	(% of RE)
United States	33,502	8.6	10.5	7.3	8.4	28.9	30.6	24.5
Europe	64,680	6.2	6.2	4.0	3.2	22.2	23.5	11.1
United Kingdom	12,323	8.5	9.4	4.9	4.8	27.2	28.9	17.7
Sweden	8,982	3.7	5.4	2.5	3.3	23.4	27.9	9.7
Poland	6,503	8.2	9.1	5.0	3.9	14.2	15.4	4.7
France	6,098	8.4	10.6	5.5	6.8	28.5	32.3	14.5
Germany	5,562	3.0	2.0	2.2	1.3	17.8	18.3	10.2
Italy	3,646	3.9	3.0	2.9	1.7	25.7	24.3	5.6
Romania	2,479	5.4	3.9	5.2	3.6	5.8	4.1	1.5
Norway	2,380	5.1	5.8	2.6	2.6	21.2	24.9	5.8
Switzerland	1,930	4.6	3.9	2.1	3.2	16.3	18.8	10.1
Finland	1,774	3.3	5.0	0.6	0.8	28.2	30.4	1.7
Spain	1,618	7.4	6.3	5.8	5.0	33.1	34.1	17.5
Greece	1,551	13.3	2.0	11.9	1.3	19.9	8.2	6.8
Denmark	1,536	5.4	6.3	3.6	1.1	25.6	26.6	13.3
Netherlands	1,386	7.1	3.8	5.4	0.0	32.2	29.8	10.1
Belgium	1,041	6.9	5.3	2.4	1.7	24.1	20.0	9.7
China	54,008	1.1	1.1	0.7	0.6	2.9	3.1	2.0
Japan	33,851	0.4	0.4	0.1	0.1	2.3	2.3	0.7
Taiwan	21,252	0.3	0.1	0.3	0.0	0.8	0.7	0.0
Korea	18,821	0.6	0.6	0.2	0.3	1.8	1.8	5.7
Australia	18,632	9.6	9.1	4.5	4.0	16.8	16.9	2.2

A Intangible capital measures

We use five measures of internally generated intangible capital. The main measure follows Peters and Taylor (2017). Because the series they distribute through WRDS ends in fiscal year 2023, we replicate their method from the raw data and use the replication to extend the series to 2025. We also replicate the measure of Iqbal et al. (2025) by translating the authors' code, and we use the measure of Ewens et al. (2025) and the organization capital of Eisfeldt and Papanikolaou (2013). This appendix describes how each measure is constructed and, for the two replications, how closely they reproduce the original series.

A.1 Peters and Taylor (2017): replication and extension

Construction. Intangible capital is the sum of knowledge capital and organization capital, each accumulated by the perpetual-inventory method from Compustat Fundamentals Annual (industrial format, all firms from 1950). Knowledge capital accumulates R&D (XRD). Organization capital accumulates 30% of SG&A, measured as XSGA minus R&D and in-process R&D (RDIP), because Compustat includes R&D in XSGA; when R&D exceeds XSGA but is below the cost of goods sold, or when XSGA is missing, we use XSGA as reported. Knowledge capital depreciates at the industry-specific rates of the Bureau of Economic Analysis, assigned by four-digit historical NAICS code, and at 15% for industries without a BEA rate and for fiscal years without a historical NAICS code (all years before 1985); organization capital depreciates at 20%. Both stocks are accumulated in real terms, $K_t = (1-\delta)\,K_{t-1}\,CPI_t/CPI_{t-1} + I_t$, with the annual average of the CPI-U, and are expressed in current dollars. Missing R&D and SG&A are set to zero, except in years in which total assets are also missing, when they are interpolated from the neighboring years. Before 1977, when Compustat's coverage of R&D is incomplete, missing R&D is set to zero if the firm reports no R&D in 1977 and is otherwise interpolated or backcast with the average R&D growth by age since the IPO. The initial stock in a firm's first Compustat record capitalizes the investment the firm made before it entered Compustat: first-year investment is grown back to the firm's founding year (from Jay Ritter's data, otherwise seven years before the IPO) at a constant pre-IPO growth rate and, for firms whose IPO precedes their first Compustat record, at age-specific growth rates between the IPO and that record.

Parameters recovered from the WRDS series. Several parameters of the WRDS series differ from, or are not stated in, the published description of the method, and we recover them from the WRDS stocks. For firms whose first Compustat record coincides with their IPO, the ratio of the initial stock to first-year investment is a geometric sum that is identical across firms with the same distance between founding and IPO, which identifies the pre-IPO

growth rates exactly: 0.3564 for R&D and 0.3473 for SG&A, against 0.348 and 0.333 in the published description. The same ratios show that firms without a founding year are backcast seven rather than eight years, and that the initial knowledge stock is computed with a 15% depreciation rate for all firms, the industry rates applying from the second year on. The inflation factor implied by the WRDS stocks equals annual-average CPI-U inflation in every year (correlation 1.0000 over 1980–2023).

Validation. Our replication reproduces the WRDS stocks within 5% for 91% of firm-years over 1950–2023 and for 94% of firm-years since 1980. For organization capital the shares are 97% and 98%, and for knowledge capital 78% and 83%. The correlation between our total intangible capital and the WRDS series is 0.999, and the median ratio of our values to the WRDS values is 1.000 in every year from 1985 on. The remaining differences come mostly from knowledge capital over 1985–1999, where about 7% of firm-years imply a BEA depreciation rate different from the one given by the historical NAICS code in the current Compustat vintage, consistent with revisions of historical industry codes since the WRDS series was built, and from the initial stocks of firms whose IPO precedes their first Compustat record. For firms that enter Compustat after 2010, our stocks match the WRDS stocks within 5% for 69% of firms in their first year, 83% in the second, 92% in the third, and more than 94% thereafter.

Extension to 2024 and 2025. Our main measure equals the WRDS series through 2023. For 2024 and 2025, we roll each firm's WRDS stocks forward with the accumulation rules above, using for knowledge capital the firm-specific depreciation rate implied by its WRDS series, and we use our own replication for firms that are not in the WRDS series. To test the procedure, we roll the WRDS stocks of 2022 forward one year: the result reproduces the WRDS stocks of 2023 within 1% for 98% of firms (median ratio 1.0000); rolling 2014 forward reproduces 2015 within 1% for 97% of firms. In 2024 (2025), the rolled-forward stocks cover 5,035 (4,724) firms, and our replication adds 633 (847) firms that are not in the WRDS series. The median growth of the stocks, 6.8% in 2024 and 6.4% in 2025, lies within the range of 2019–2023 (6–15%).

A.2 Iqbal, Rajgopal, Srivastava, and Zhao (2025)

Iqbal et al. (2025) estimate, for each Fama–French 48 industry and year, the share of R&D and of operational SG&A that is investment and the useful life of that investment, from the relation between current expenses and future sales. Because their data end in 2022, we translate their SAS replication package to Stata line by line and do not tune the translation to match their data. Operational SG&A is XSGA (XAGT when XSGA is missing) minus

R&D and in-process R&D, floored at zero. For each industry and year, R&D (operational SG&A) relative to average assets is regressed on current sales and up to seven (five) years of future sales, relative to average assets, over a rolling window of firm-years for which all seven (five) future years of sales are available. Each regression is refit after excluding observations with absolute studentized residuals above three, and the specification with the highest adjusted R^2 determines the useful life, that is, the number of future years of sales. The investment share is the industry mean of the fitted investment component relative to the expense, capped to $[0, 1]$. Two further rounds re-estimate the regressions with average assets that include the capital from the previous round, and the final estimates are winsorized at the 1st and 99th percentiles by industry. Capital is the investment share of past expenses, amortized straight-line over the useful life and computed with the industry estimates of the previous year.

Run on Compustat data through 2022, our translation yields samples about 2% larger than those reported in the authors' log (276,324 against 271,863 firm-years in the base sample), consistent with the newer Compustat vintage, and matches 96% of the firm-years in the authors' shared data (1980–2022). The correlation with their values is 0.93 for capitalized operational SG&A and 0.88 for capitalized R&D, the rank correlation of total intangible capital is 0.87, and the median ratio of our values to theirs is 1.00. The match is not closer because the useful life selected by the highest adjusted R^2 is unstable: in the authors' own estimates, an industry keeps the same useful life from one year to the next only 49% of the time (50% in ours), although consecutive estimation windows share most of their data, so that a 2% difference in the sample changes the selected life in many industry-years. The distributions of the estimates are nevertheless very similar: the average useful life of operational SG&A is 3.3 years in the authors' estimates and 3.5 years in ours, and the average investment share 0.69 and 0.66. Two features of the method matter for its use in our tests. First, because the final winsorization uses all years and the estimation windows at the end of the sample are short, the estimates of the last two to three years are revised substantially when new data arrive (by a median of 12–20% for 2020–2022 when we add the data through 2025), so the measure is provisional for the most recent years. Second, the regressions use sales up to seven years after the balance-sheet date, so the measure contains information that is not available when portfolios are formed; we therefore use it only as a robustness check.

A.3 Ewens, Peters, and Wang (2025) and Eisfeldt and Papanikolaou (2013)

For Ewens et al. (2025), we use the intangible capital series the authors distribute through intangiblesdata.org (the version of September 2026), defined as the sum of their knowledge capital and organization capital, matched by GVKEY and fiscal year. For Eisfeldt and Papanikolaou (2013), we compute organization capital from Compustat as $OC_t = (1 - \delta)\, OC_{t-1} + SG\&A_t$, with all of SG&A (XSGA, set to zero when missing or negative) treated as investment, a depreciation rate of 20% as in Eisfeldt et al. (2022), and an initial stock in the firm's first Compustat record of $(1 - \delta)\, SG\&A/(g + \delta) + SG\&A$ with $g = 10\%$; unlike the original measure, which is computed from deflated SG&A, ours is accumulated in nominal terms. Figure A1 and Figure A2 in Appendix B show the anatomy of red equity, and Table A4 the value-factor results, for each measure.

Table A1: Variable definitions

This table defines the main variables used in the paper. Compustat items are in parentheses.

Variable	Definition
Age	Years since the IPO: fiscal year minus IPO year, set to zero before the IPO. The IPO year is the year of the firm's first offering in Jay Ritter's IPO database (matched by PERMNO); if not available, the year of the Compustat IPO date (IPODATE); if that is also missing, the first fiscal year in Compustat. Used in years in the regressions of RE status and as $\log(1 + Age)$ in the leverage regressions.
Annual_Return	Compounded monthly stock return over the fiscal year. Data from CRSP.
Default probability	Naive default probability of Bharath and Shumway (2008): one-year Merton-type probability using market equity, debt (short-term debt plus half of long-term debt), equity volatility (12 monthly returns), and the past annual return.
Delisted	= 1 if the stock has a performance-related CRSP delisting within 24 months after the fiscal year end, and 0 otherwise.
Distressed	= 1 if at least two of the following hold: Altman Z-score below 1.81 (Altman, 1968), default probability above 10%, EBITDA below interest expense.
Distressed (int.-adj.)	As Distressed, but with the Altman Z-score and interest coverage adjusted for expensed intangible investment: intangible investment is added back to EBIT and EBITDA, and intangible capital to retained earnings and total assets.
Dividend	Log of dividends. Data from Compustat.
Dividend_over_Assets	The ratio of dividends to total assets. Data from Compustat.
Ind_Leverage	Median leverage of firms in the same two-digit SIC industry in the year.
Intangible_Investments	Log of intangible investment, $Intan = 0.3 \times$ SG&A + R&D following Peters and Taylor (2017). Data from Compustat.
Intan_over_Assets	The ratio of intangible investment, $Intan$, to total assets. Data from Compustat.
Leverage	The ratio of total debt (DLTT + DLC) to total assets (AT). Data from Compustat.
Lifetime earnings	Retained earnings (RE) in the first fiscal year in which the firm reports them, plus common dividends (DVC) paid in earlier fiscal years, plus the sum of net income (NI) since that year. For firms that never report retained earnings, the sum of net income after the first Compustat record. Data from Compustat Fundamentals Annual.
LossMaking	= 1 if net income is negative, and 0 otherwise. Data from Compustat.
Payout_over_Assets	The ratio of payouts, common dividends (DVC) plus repurchases (PRSTKC, set to zero when missing), to total assets. Data from Compustat. Dividends only before 1971.
Profitability	The ratio of operating income before depreciation (OIBDP) to total assets. Data from Compustat.
RE type	Source of negative equity for RE firms: payout-driven if lifetime earnings are non-negative; write-down-driven if lifetime earnings are negative but non-negative after adding back cumulative write-downs; operating-loss-driven otherwise. Loss-driven combines the last two.
RedEquity	= 1 if common equity is negative and 0 otherwise. Data from Compustat.
Repurchase	Log of repurchases. Data from Compustat. Missing before 1971.
Repurchase_over_Assets	The ratio of repurchases to total assets. Data from Compustat. Missing before 1971.
Size	Log of market capitalization. Data from CRSP.
Tangibility	Tangible fixed assets (PPENT) over total assets. Data from Compustat.
Tax	The ratio of total income taxes (TXT) to total assets. Data from Compustat.
TobinQ	The ratio of market capitalization plus preferred stock plus total debt to total assets. Market capitalization from CRSP and the rest from Compustat.
Writedown	Log of write-downs. Data from Compustat. Missing before 1995.
Writedown_over_Assets	The ratio of write-downs and impairments to total assets. Data from Compustat. Missing before 1995.

B Additional results

Table A2: Determinants of red equity: Logit regressions

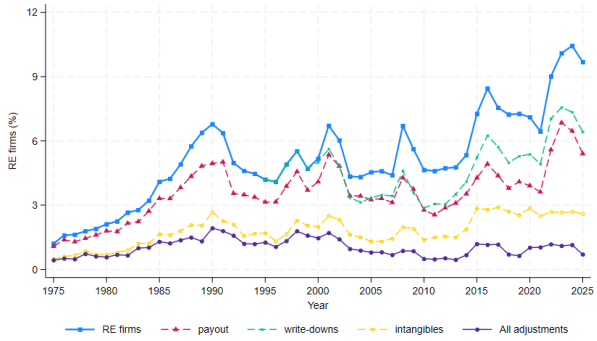
This table repeats Table 4 with logit instead of linear probability models. The dependent variable is a binary indicator equal to 1 if a firm has negative equity in a given year, and 0 otherwise. The table reports average marginal effects, multiplied by 100 so that they are in percentage points and comparable with the estimates of Table 4; for the indicator variables, they are the average derivatives of the predicted probability. All non-binary independent variables are standardized each year to have mean 0 and standard deviation 1. Columns 5 and 6 include year fixed effects, and Column 6 also industry (Fama–French 49) fixed effects. Standard errors, clustered at the firm level and computed with the delta method, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. All variables are defined in Table A1.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	RedEquity					
LossMaking	6.68***	6.97***	6.41***	2.22***	1.83***	1.72***
	(0.19)	(0.19)	(0.28)	(0.19)	(0.19)	(0.19)
Profitability	-0.73***	-0.81***	-1.08***	-0.25**	-0.44***	-0.45***
	(0.08)	(0.07)	(0.16)	(0.12)	(0.13)	(0.12)
Intan_over_Assets	0.44***	0.39***	0.63***	1.83***	1.74***	2.07***
	(0.08)	(0.08)	(0.16)	(0.13)	(0.14)	(0.14)
Payout_over_Assets		0.71***	1.23***	0.82***	0.79***	0.71***
		(0.09)	(0.15)	(0.08)	(0.08)	(0.08)
Writedown_over_Assets			0.42***	0.18***	0.18***	0.11***
			(0.06)	(0.04)	(0.04)	(0.04)
Size				-1.42***	-1.49***	-1.59***
				(0.14)	(0.14)	(0.14)
Leverage				5.74***	5.73***	5.65***
				(0.12)	(0.11)	(0.11)
Age				0.65***	0.70***	0.76***
				(0.16)	(0.16)	(0.16)
TobinQ				1.28***	1.21***	1.28***
				(0.09)	(0.09)	(0.09)
Observations	200,369	200,369	101,518	101,097	101,097	101,097
Year FE	No	No	No	No	Yes	Yes
Industry FE	No	No	No	No	No	Yes

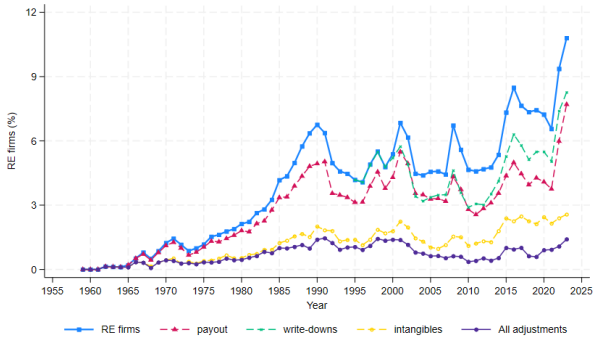
Figure A1: Anatomy of red equity: other intangible capital measures

This figure repeats Figure 12 with the other measures of intangible capital: (a) Ewens et al. (2025); (b) Peters and Taylor (2017) as available on WRDS, which ends in 2023; (c) the organization capital of Eisfeldt and Papanikolaou (2013); (d) Iqbal et al. (2025), using our replication of their code. Each panel plots the share of red equity firms among all firms and the share that would remain red equity after adding back cumulative payouts, adding back cumulative write-downs and impairments, adding intangible capital, and all three adjustments together. Each panel uses only the firm-years for which the measure is available, so that all lines compare the same firms; because the measures cover different firms and years, the red equity line differs slightly across panels. Figure A2 uses all firm-years instead.

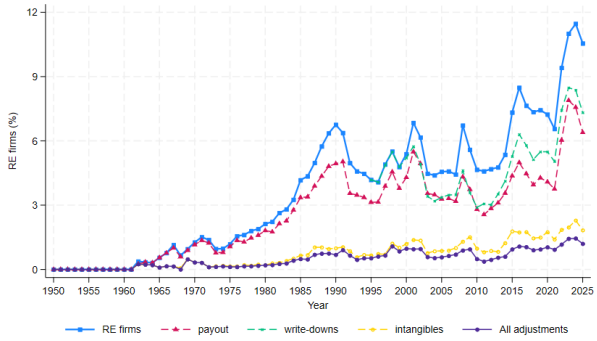
(a) Ewens, Peters, and Wang



(b) Peters and Taylor (WRDS)



(c) Eisfeldt and Papanikolaou



(d) Iqbal et al.

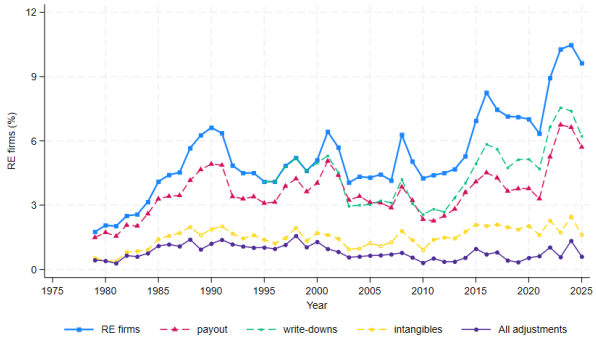
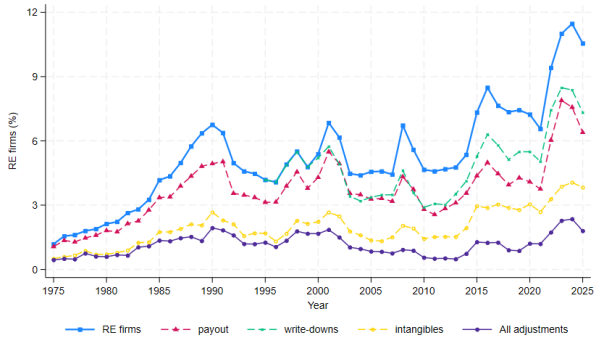


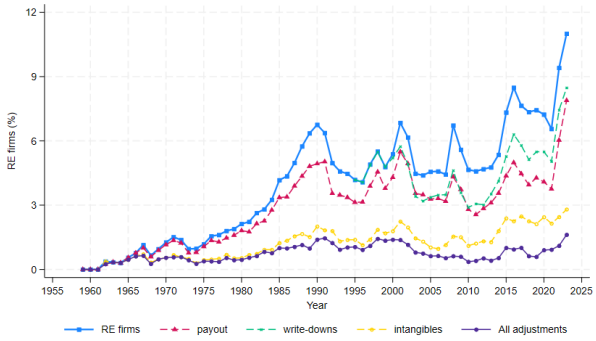
Figure A2: Anatomy of red equity: other intangible capital measures, all firm-years

This figure repeats Figure A1 on all firm-years: a firm-year without intangible capital in a measure is kept, with intangible capital set to zero. The red equity line is therefore the same in all panels (apart from the years each measure covers), but the intangibles adjustment is understated for firms that a measure does not cover. Panels: (a) Ewens et al. (2025); (b) Peters and Taylor (2017) as available on WRDS, which ends in 2023; (c) the organization capital of Eisfeldt and Papanikolaou (2013); (d) Iqbal et al. (2025).

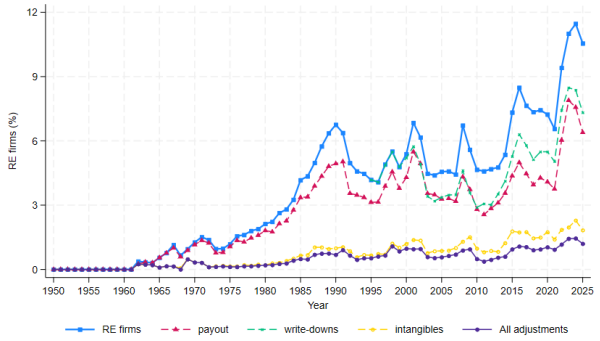
(a) Ewens, Peters, and Wang



(b) Peters and Taylor (WRDS)



(c) Eisfeldt and Papanikolaou



(d) Iqbal et al.

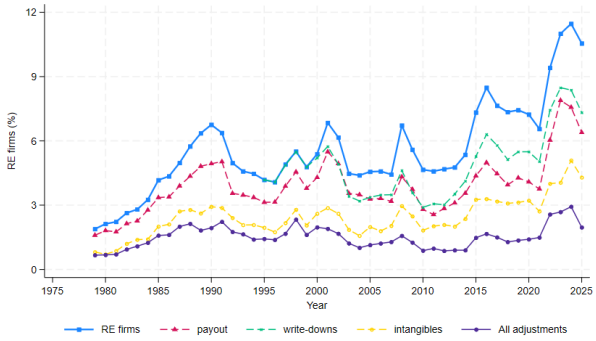


Figure A3: Negative equity after deducting goodwill and other intangible assets

This figure plots the share of firms (panel a) and of total market capitalization (panel b) with negative equity when different intangible assets on the balance sheet are deducted from common equity: all intangible assets (tangible equity, as in Figure 2), goodwill only, intangible assets other than goodwill only, and none (current GAAP). The series use the same firm-years as Figure 2 and start in 1988, the first fiscal year in which Compustat reports goodwill separately from other intangible assets.

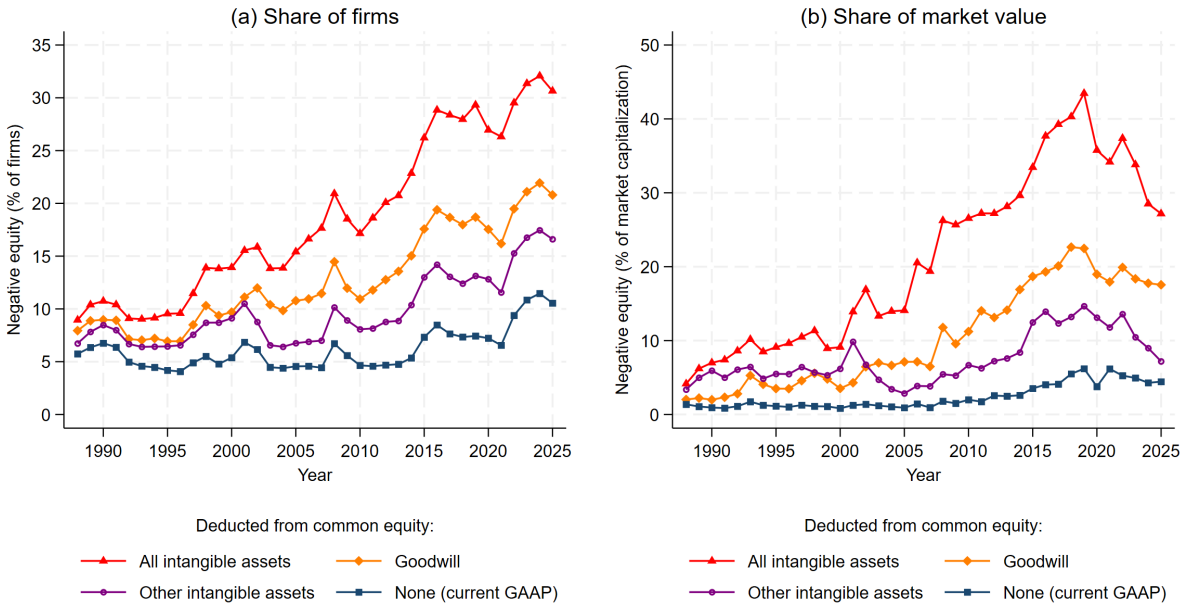


Figure A4: Including red equity firms in intangible-adjusted HML, by start of the sample: controlling for momentum

This figure repeats Figure 16 with the momentum factor of Carhart (1997) added to the factor regression. It plots the difference between HML_INT inc RE and HML_INT ex RE in average return, CAPM alpha, and FF + momentum alpha, in percent per year, for samples that start in July of the year shown and end with the data. The FF + momentum alpha is the intercept of a regression of the monthly difference on the market, size, profitability, and investment factors of Fama and French (2015) and the momentum factor. The shaded area is the 95% confidence interval of the FF + momentum alpha, based on Newey–West standard errors with one lag.

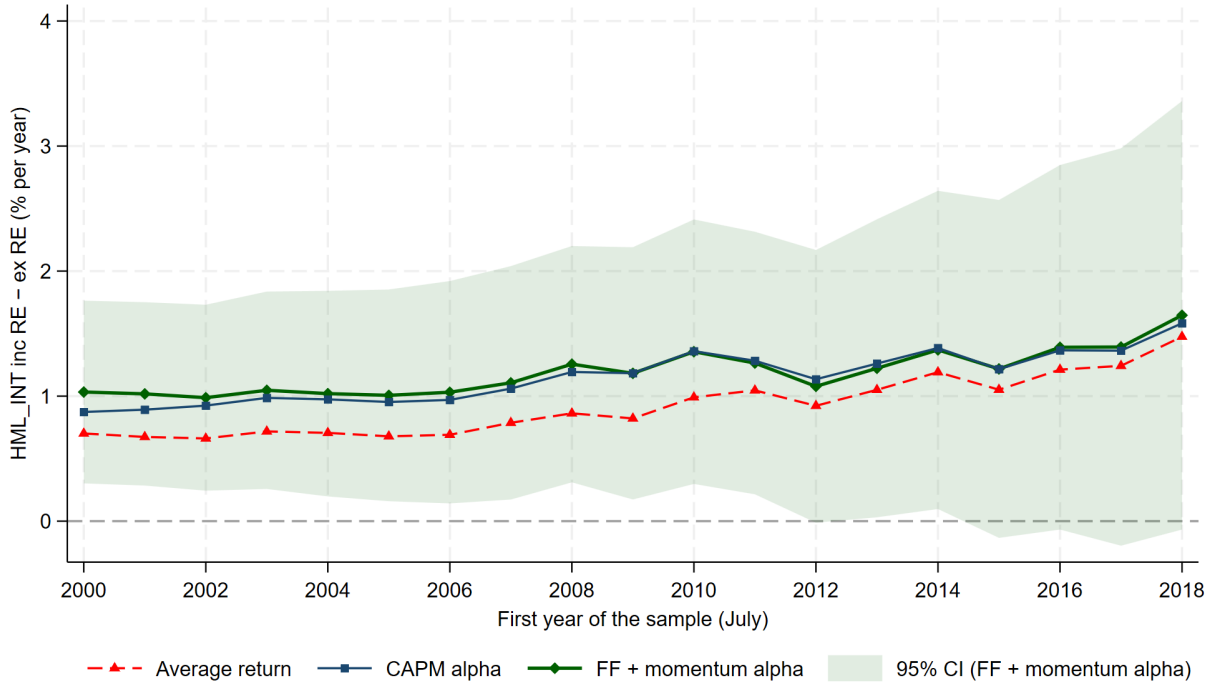


Table A3: Value relevance of reported and intangible-adjusted book value: scaled by total assets

This table repeats Table 11 with market value, book value and earnings scaled by total assets instead of per share. Both panels use the firm-years with intangible capital in the current and the previous year, 1975–2025. All regressions include year and industry (Fama-French 49) fixed effects. Variables are winsorized at the 1% level in each tail. Standard errors, clustered at the firm level, are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The last two rows of each panel report the p-values of tests that the book-value coefficient of payout-driven RE firms equals that of positive-equity firms, (2) = (1), and that of loss-driven RE firms, (2) = (3), from a pooled regression in which book value, earnings, and the year and industry fixed effects are interacted with indicators for the two types of RE firms, so that its coefficients equal those of the separate regressions.

	(1)	(2)	(3)
	Positive	RE firms	
	book equity	Payout-driven	Loss-driven
Panel A: reported book value and earnings			
Book value	1.80***	-2.11***	-0.43***
	(0.04)	(0.54)	(0.16)
Earnings	-0.62***	4.07***	-1.56***
	(0.06)	(0.70)	(0.10)
Observations	166,260	1,576	7,620
R-squared	0.22	0.44	0.28
p-value, book value: (2) = (1)		< 0.001	
p-value, book value: (2) = (3)		0.002	
Panel B: intangible-adjusted book value and earnings			
Adjusted book value	0.54***	0.74***	0.34***
	(0.02)	(0.18)	(0.04)
Adjusted earnings	0.68***	4.96***	-1.24***
	(0.06)	(0.72)	(0.11)
Observations	166,260	1,576	7,620
R-squared	0.19	0.45	0.27
p-value, book value: (2) = (1)		0.265	
p-value, book value: (2) = (3)		0.027	

Table A4: Including red equity firms in intangible-adjusted HML: all intangible capital measures

This table reports the difference between HML_INT inc RE and HML_INT ex RE (column (5) of Table 12) when intangible-adjusted book equity is computed with each intangible capital measure: Peters and Taylor (2017), extended to 2025 with our replication (PT ext.); the Peters and Taylor (2017) series on WRDS (PT WRDS); Ewens, Peters, and Wang (2025); Eisfeldt and Papanikolaou (2013) organization capital (EP); and Iqbal et al. (2025) with our replication of their code. The Iqbal et al. (2025) estimation uses firms' sales up to seven years after the balance sheet date, so this measure contains information that was not available when the portfolios were formed. Each measure is used over its own sample, which starts when the measure covers enough firms and ends with its data. Differences are annualized and in percent; FF alpha is the intercept of a regression on the market, size, profitability, and investment factors. Newey–West t-statistics with one lag are in parentheses.

		(1)	(2)	(3)
Measure	Sample	Average return	CAPM alpha	FF alpha
Panel A: full sample				
PT ext.	1970–2026	0.47	0.57	0.75
		(2.55)	(3.24)	(3.89)
PT WRDS	1970–2025	0.48	0.58	0.72
		(2.65)	(3.34)	(3.78)
Ewens et al.	1976–2026	0.21	0.31	0.40
		(0.79)	(1.19)	(1.48)
EP	1970–2026	0.30	0.32	0.46
		(2.12)	(2.29)	(2.97)
Iqbal et al.	1980–2026	0.28	0.40	0.71
		(0.91)	(1.28)	(2.03)
Panel B: from July 2009				
PT ext.	2009–2026	0.82	1.18	1.30
		(1.62)	(2.45)	(2.63)
PT WRDS	2009–2025	0.84	1.23	1.35
		(1.63)	(2.51)	(2.61)
Ewens et al.	2009–2026	0.32	0.74	0.73
		(0.53)	(1.31)	(1.32)
EP	2009–2026	0.58	0.67	0.84
		(1.57)	(1.85)	(2.36)
Iqbal et al.	2009–2026	0.60	1.27	1.22
		(0.96)	(1.97)	(1.85)

Table A5: Red equity in other countries

This table reports the share of listed firms with negative common equity (RE firms), the same share excluding the smallest 20% of firms by total assets in each country and year, the share that would have negative equity if all intangible assets on the balance sheet were expensed (tangible equity, common equity minus intangible assets including goodwill, below zero), and the share of RE firm-years that are payout-driven (lifetime earnings non-negative). The last column is left blank for countries with fewer than 30 RE firm-years over 2015–2025, for which the share is not meaningful. Data for countries other than the United States come from Compustat Global (industrial format, consolidated statements, exchange-listed firms, excluding utilities, banks, and insurance companies, as in the US sample); common equity is reported under each firm's accounting standards. The table reports the non-European countries with at least 1,000 listed firm-years over 2015–2025 that are not shown in Table 14. India is excluded because of a break in its series in 2011, when most Indian firms with negative equity had positive equity a year earlier; from fiscal 2011–12, India's Revised Schedule VI required accumulated losses previously shown on the assets side of the balance sheet to be deducted from shareholders' funds. Firms incorporated in Bermuda, the Cayman Islands, and the British Virgin Islands are also excluded. Data for the United States come from the main sample. Averages are pooled over 2015–2025.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Firm-years	RE firms (%)		RE firms excl. smallest 20% (%)		Negative tangible equity (%)		Payout-driven
	2015–2025	2015–2025	2025	2015–2025	2025	2015–2025	2025	(% of RE)
Malaysia	9,796	1.7	1.4	1.1	1.0	3.7	2.6	5.3
Thailand	7,570	1.4	1.6	1.2	1.6	4.5	4.3	7.5
Indonesia	6,932	5.6	4.3	4.3	4.2	7.0	5.2	1.5
Singapore	5,564	5.8	6.7	2.3	1.8	7.7	9.2	7.4
Vietnam	5,258	3.0	2.8	2.3	1.5	3.8	3.2	8.2
Turkey	4,130	2.8	1.4	2.7	1.1	5.5	5.4	2.6
Israel	4,121	7.9	6.7	2.2	0.3	13.8	11.1	4.0
Pakistan	3,816	8.3	7.8	5.2	3.9	8.8	9.2	1.9
Brazil	3,051	15.1	12.8	8.8	8.0	27.1	23.9	2.6
Saudi Arabia	2,337	0.8	1.1	0.7	1.3	3.0	3.6	
Bangladesh	2,207	3.9	4.6	0.9	1.4	4.1	4.7	15.3
Sri Lanka	2,178	2.8	4.7	1.7	3.2	3.7	4.2	3.2
Philippines	1,896	4.2	5.9	2.1	2.5	9.8	8.8	5.1
South Africa	1,891	3.0	0.8	1.6	1.0	8.3	3.8	33.9
Hong Kong	1,573	5.4	6.1	2.1	2.9	9.9	10.2	5.9
Egypt	1,554	5.4	2.3	5.1	2.9	6.0	1.6	28.6
Russia	1,492	10.6	10.7	11.2	10.0	17.3	17.8	15.8
Chile	1,274	1.3	2.0	1.2	2.5	4.2	3.1	
New Zealand	1,245	3.0	2.1	1.0	0.0	15.0	14.0	8.1
Mexico	1,038	1.2	0.0	0.6	0.0	15.0	14.1	
Nigeria	1,024	11.5	12.5	8.3	9.4	13.3	13.0	16.1
Jordan	1,010	4.1	1.5	3.5	1.9	6.7	4.7	7.3